

Nuclear gluonometry with a tensor-polarized ^6Li beam at the Electron-Ion Collider: projected inclusive and coherent $\cos 2\phi$ measurements

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Abstract. The double-helicity-flip structure function $\Delta(x, Q^2)$ of a spin-1 target is generated at leading twist by gluons alone and has never been measured. We project its measurement at the Electron-Ion Collider with a transversely tensor-polarized ^6Li beam, in inclusive deep inelastic scattering, where only the scattered electron is detected, and in coherent diffraction with the ^6Li recoiling intact. With the leading-order bag-model moment of Sather and Schmidt ported to ^6Li as a labelled scenario, the inclusive $\cos 2\phi$ amplitudes at the four most sensitive (x, Q^2) bins, $x = 0.011\text{--}0.14$ and $Q^2 = 1.1\text{--}14 \text{ GeV}^2$, are $(0.44\text{--}0.95) \times 10^{-2}$ at $P_{zz} = 0.6$, against single-state truth-level statistical errors of $(1.4\text{--}4.5) \times 10^{-4}$ in one EIC year (10 fb^{-1} per nucleon): a $21\text{--}43\sigma$ measurement per bin whose x dependence discriminates between moment-constrained interpretations, and an extraction of $x\Delta(x, Q^2)$ itself to $2\text{--}9\%$ in the best bins. At the reconstructed level, with the spin-state-sorted ratio of $m = \pm 1$ -rich and $m = 0$ -rich bunches and a hadronic-method y from a PYTHIA 8 final state, the errors are $(0.9\text{--}3.0) \times 10^{-4}$ and the fit closes on the response-folded truth. The coherent channel carries the same modulation with two separable mechanisms — a nuclear-shape term anchored on the polarized-deuteron colour-glass calculation, predicted to flip sign for ^6Li , and a flat exotic-gluon term — and ^6Li 's anomalously small quadrupole moment makes it a null test. Its intact recoil keeps the beam rigidity and, with the Yellow Report's beam divergences, cannot be tagged at IP6 with any published optics; the channel defines a far-forward requirement — a dedicated tagging optics or the second interaction region's secondary focus — and for a reference tag with a 0.20 GeV near-beam envelope at the high-acceptance luminosity — which no IP6 optics provides — one year would yield 1.7×10^7 events and a 5σ sensitivity to an amplitude of 0.9% per unit P_{zz} , against a shape term of 6% and an exotic-gluon term of $0.3\text{--}1\%$ on the same footing. A tagging optics at IP6 — the horizontal β^* raised $46\text{--}164\times$ with the pots following the envelope, the vertical plane untouched — recovers a $25\text{--}37\%$ tag at $1/6.8\text{--}1/12.8$ of the luminosity: $2.4\text{--}6.1 \times 10^6$ tagged events per year, an $8\text{--}11\sigma$ measurement of the shape term per year and 5σ on a 1% exotic-gluon term in $3\text{--}6$ years, both on the counting floor of the best (x, Q^2) super-bin alone (the seven- $|t|$ -bin fit of the whole tagged sample measures the flat term to $\delta a_e = 0.07\text{--}0.12\%$ per unit P_{zz} per year), $4\text{--}10\times$ below what a second interaction region's secondary focus would deliver at the same luminosity.

1 · Introduction

A spin-1 target can absorb two units of photon helicity in the forward Compton amplitude. The structure function that measures this, $\Delta(x, Q^2)$, receives no leading-twist contribution from quarks or from bound spin- $1/2$ nucleons, so in a nucleus it is direct evidence for glue that is not inherited from the individual nucleons — "exotic glue" in the language of Jaffe and Manohar's nuclear gluonometry [1,2]. It has never been measured for any target; the one prior experimental document is a JLab letter of intent for a ^{14}N target, without a sensitivity projection [3]. The only absolute normalizations available are moment estimates: a bag-model calculation for the spin- $3/2$ Δ^{++} baryon [4], maximal-transversity scenarios for proton–deuteron Drell–Yan [5], and lattice-QCD matrix elements for the ϕ meson [6] and the deuteron [7]. The EPIOS white paper on polarized ion beams at the EIC names spin-1 ^6Li among the nuclei where a discovery is most likely [8], and the companion physics case [9] argues why ^6Li — a polarized deuteron bound to an inert α particle, with $A/Z = 2$ and deuteron-like spin dynamics in the ring — is the practical choice.

This paper presents the first numerical projection of a Δ measurement for any target. Two channels are considered. In inclusive DIS on a transversely tensor-polarized ^6Li beam the scattered electron's azimuth about the alignment axis acquires a $\cos 2\phi$ modulation proportional to Δ . In coherent diffraction, $e^+{}^6\text{Li} \rightarrow e^+ X^+{}^6\text{Li}(\text{g.s.})$, the nucleus stays in its

ground state and the modulation of the tagged yield mixes the gluonic term with a nuclear-shape term that the polarized-deuteron calculation of Mäntysaari et al. [10] predicts — and ${}^6\text{Li}$, with a quadrupole moment fifty times smaller than ${}^7\text{Li}$'s, separates the two. Section 2 states the observable, Section 3 the simulation framework, Section 4 the Δ model, Section 5 the inclusive projections at truth and reconstructed level, Section 6 the coherent channel with the far-forward requirement it imposes, Section 7 the systematics and assumptions.

2 • The observable

For unpolarized electrons on a spin-1 target whose spin axis makes angle θ_m with the beam, the inclusive cross section per spin projection m is [1,2]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3} a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3} a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_m \Delta(x, Q^2) \cos 2\phi \right]$$

$$a_m = \frac{1}{4} c_m (3 \cos^2 \theta_m - 1), \quad c_m = 3|\lambda_m| - 2 \rightarrow (1, -2, 1)$$

with ϕ the azimuth of the lepton plane about the virtual photon, measured from the transverse projection of the spin axis. Averaged over a fill, c_m becomes the tensor polarization $P_{zz} = p_+ + p_- - 2p_0$. The b_1, b_2 terms are written above in the sign of this programme's own transcription of [2], the form the companion physics case [9] displays as well; since 2026-08-29 the simulation carries the opposite one — the Hoodbhoy–Jaffe–Manohar convention as applied by HERMES and by Cosyn et al. [11], $A_{zz} = -(2/3) b_1/F_1$ for a spin axis along the photon, which is the convention every published b_1 is quoted in. It is one constant, and it affects only the sign of A_{zz} relative to b_1 , the constant term of the spin-state ratio of §3.3 and the sign of the $O(\gamma^2)$ leakage below; nothing in the Δ sector itself. It reaches a magnitude only through the 10^{-3} of the rate the b_1 term carries, which moves one printed digit here — the third sweet spot's one-year error of $\$5.1, 2.8 \rightarrow 2.7 \times 10^{-4}$. At leading order Δ is generated by the gluon transversity distribution δG through the $O(\alpha_s)$ photon–gluon convolution

$$\Delta(x, Q^2) = -\frac{\alpha_s(Q^2)}{2\pi} \left(\sum_q e_q^2 \right) x^2 \int_x^1 \frac{dy}{y^3} \delta G(y, Q^2)$$

in the sign convention of Ref. [6]. For a transverse spin axis ($\theta_m = 90^\circ$) the measured amplitude and its inversion to the structure function are

$$A \cos 2\phi = -\frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2, \quad \hat{\Delta} = -\hat{A} \frac{y^2 D_\phi}{1-y}$$

Three properties of this relation shape the analysis. First, with $F_2 = 2x(1+R)F_1$ the inversion becomes $\Delta/F_1 = -\hat{A}[2(1+R) + y^2/(1-y)]$, so at the $y \lesssim 0.03$ of the sensitive region F_2 cancels and $R = \sigma_L/\sigma_T$ is the only structure-function input that reaches Δ/F_1 . Second, the amplitude is exactly linear in Δ (§5.3). Third, the target-mass corrections are small: at the four bins of §3.4, $\gamma^2 = 4M^2 x^2/Q^2 \leq 0.0057$ with the free-nucleon $M = 0.9383$ GeV that a per-nucleon x requires (the bound-nucleon mass would lower it by 1.0%), so the approximation set of the simulation — $g_2 = g_2^{WW}$, and until 2026-08-29 $A_{||} = D \cdot g_1/F_1$ with the γ^2 -suppressed term dropped — is adequate. The exact longitudinal kernel $A_{||} = D_V(A_1 + \eta A_2)$ is now the default (`InclusiveKernel(target_mass=True)` ; `target_mass=False` restores the massless limit); at the $y \lesssim 0.03$ of these bins it collapses to a multiplicative $(1 + \gamma^2)$ on $A_{||}$, independent of g_2 — $\leq 0.6\%$ here, $\leq 0.5\%$ at the top configuration and $\leq 2.5\%$ at the low one, whose $Q^2 = 1.14$ GeV² spot sits at $x = 0.089$ — and the $W^2 \geq 10$ GeV² cut bounds it by $\gamma^2 \leq M^2/(W_{\min}^2 - M^2) = 0.097$ anywhere in the acceptance. No Δ number reported here is built from $A_{||}$. One finite- γ effect is physical rather than a labelling matter: the tensor-polarized b_1 – b_4 terms feed the $\cos 2\phi$ amplitude at $O(\gamma^2)$, because a spin axis transverse to the beam is not transverse to the photon. It is now computed rather than bounded — the exact kernel of Cosyn et al. Eqs. (9), (10), (14), (17) and (24) is in the generator (`InclusiveKernel(tensor_gamma=True)` , off by default so that the massless path every published number was made on is untouched) and `evgen/scripts/tensor_gamma_leakage.py` measures the leakage at the twelve sweet spots of the three configurations. Two of its three channels very nearly cancel — the leading-twist rate term and the twist-3 Eq. (17d) one stand as +3 and –3 against the twist-4 Eq. (17e) term, exactly so as γ^2 and y go to zero — so what survives is Eq. (17e) almost alone, an effective Δ of $(0.14\text{--}0.16) \gamma^2 b_1$, worth at most -0.033% of the amplitude at the four bins here,

-0.027% at the top configuration and -0.109% at the low one, whose γ^2 is four times larger. The sign is the one that helps: for a positive b_1 the leakage opposes the amplitude of a negative Δ rather than faking one, and what carries it is not b_1 but b_2 — 99.96% of the leakage is the b_2 term, and $b_2 = 2xb_1$ at leading twist — in the same combination that the ϕ' -independent constant κ of the ratio fit (Report 2 §4.3) measures, so it is subtractable in situ from the same fills rather than against a separate A_{zz} run: the correction is $-L(x, Q^2, y) D_\phi \hat{\kappa}$ with L a pure kinematic ratio, and what it leaves is the unmeasured b_3, b_4 band, 0.6 of the leakage — $\leq 0.020\%$ of the amplitude here at the truth level (0.0014 – 0.018% at the four reconstructed bins, Report 2 §4.1) and $\leq 0.066\%$ at 5×40.8 .

The size of Δ is unknown to within an order of magnitude. The bag-model moment of Sather and Schmidt, $\int x \Delta dx = -0.012 \alpha_s$ with its Q^2 evolution but no x dependence [4], puts the asymmetry at the per-mille level; the explicit $\alpha_s(Q^2)$ is the leading-order Wilson coefficient of the photon–gluon box, so the $O(\alpha_s)$ suppression relative to F_1 is model-independent while -0.012 is the bag estimate. Lattice QCD adds two results, both at unphysical pion masses: the bare ϕ -meson matrix element $A_2 = 0.23(2)(5)$ at $m_\pi = 450$ MeV [6], which converts to $\int x \Delta|_\phi \approx -0.006 \alpha_s$ (our conversion, unrenormalized), and the NPLQCD deuteron δG moment of $-0.010(3)$, bare at $m_\pi \approx 806$ MeV [7] — a transversity-to-momentum ratio roughly ten times below the ϕ 's, implying the opposite sign for the deuteron's Δ moment. Percent-level asymmetries arise only in the maximal-transversity ($\Delta_{Tg} = \Delta g$) Drell–Yan scenarios their authors call optimistic [5]. Nothing exists for ${}^6\text{Li}$ or any $A > 2$ nucleus beyond a $\Delta\Delta$ -isobar estimate for the deuteron [12], so the literature brackets the nuclear Δ in both directions: suppressed, as in the NPLQCD deuteron, or enhanced by binding in analogy with the EMC effect [8]. The projections therefore carry the moment-constrained model of §4 together with the scenario band $\Delta/F_1 \in 10^{-3}$ – 10^{-2} .

3 • Simulation framework

3.1 • Beams, polarization and luminosity

EIC ions are γ -matched to the proton rather than rigidity-scaled, because the two rings must share a revolution period, so the ${}^6\text{Li}$ menu is 40.8 GeV/u and 99.5–137.5 GeV/u, the top value being the rigidity cap $275 \text{ GeV} \times Z/A$ [8,13]. The projections are made at the mid configuration, $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV/u})$, $\sqrt{s_{NN}} = 63 \text{ GeV}$, with the low (5×40.8) and top (18×137.5) configurations used where the hadronic reconstruction or the far-forward acceptance depends on the energy. The in-ring tensor polarization is a placeholder $P_{zz} = 0.6$ (the ANL source targets $P_{zz} \geq 0.8$ [9]; transport to the ring is an open accelerator question, and every statistical error scales as $1/P_{zz}$), the alignment axis is the vertical — the stable spin direction of the arcs, assumed to reach the interaction point unrotated, which is one of the three demands §8 makes on the machine — and the electrons are unpolarized. Luminosity is quoted per nucleon: $10 \text{ fb}^{-1}/u$, "one EIC year" in the Yellow Report's accounting (30 weeks at $10^{33} \text{ cm}^{-2} \text{ s}^{-1}$ with 60% efficiency [13]), and $100 \text{ fb}^{-1}/u$ for the ten-year programme, all of it in the transverse-tensor fill. No lithium luminosity has been published; the yields scale linearly with whatever the machine delivers. The inclusive and coherent measurements want different far-forward optics — the Yellow Report divergence for the first, the tagging optics of §6.1 at 1/6.8 to 1/12.8 of the luminosity for the second — and each projection assumes the full luminosity of the year in its own, so the two reaches are alternatives rather than a sum; the share the run plan actually gives each is a programme decision, priced in plans/07 WP2, and every statistical error below scales as one over the square root of it, while a luminosity quoted as a reach in fb^{-1}/u stands where it is and the years needed to accumulate it grow as one over the share. What does not follow that law is anything in an error bar that is not statistics: the full one-year Δ bars of Report 2 [16] carry the spread over unfolding priors, which no luminosity shrinks, and it has to be separated out before a share is applied.

3.2 • Generator and structure functions

No public generator produces polarized $e+A$ events for $A > 1$: BeAGLE is unpolarized, DJANGO and PEPSI are nucleon-level and longitudinal-only. The generator used here, polligen, implements the missing component — the spin-density matrix $\otimes$ structure-function master formula with spin-correlated sampling — and takes the hadronic final state from a PYTHIA 8 production of 8 M events over the three beam configurations. It is validated against the analytic asymmetry formulas and by pseudo-experiments, including unbiased $\cos 2\phi$ recovery under ϕ -acceptance holes; a full-

luminosity pseudo-experiment draws each ϕ -bin count from a Poisson distribution with the exact expected yield, which is statistically identical to event-level sampling.

Structure functions are toy parameterizations with PDF-grid backends (CT18NLO, EPPS21, NNPDFpol11) behind one interface, so rates hold to a factor ≈ 1.5 (the toy F_2 is within $\pm 37\%$ of CT18 over the acceptance) and asymmetry shapes are scenarios by construction. $R = \sigma_L/\sigma_T$ is a placeholder of the right magnitude, 0.14–0.18 at the four bins of §3.4, in every projection below; the published SLAC/E143 R1998 world fit [14], all three of its functional forms, sits behind the same hook, and switching to it moves Δ/F_1 by +18, +18, +8 and –4% at the four bins, with a further 0.5–6% from the spread of the fit forms — a correction to apply rather than a band, and the systematic on every Δ quoted below until it is.

3.3 • Estimators

The figures use a single spin state: a binned least-squares fit of $N(\phi') = C(1 + A \cos 2\phi')$ rather than the $2\langle \cos 2\phi' \rangle$ moment, which realistic ϕ holes bias by more than the signal while the fit stays unbiased, with $\delta A = \sqrt{(2/N)/P_{zz}}$ validated by pseudo-experiments and no $\cos \phi'$ term, since the m-symmetric tensor fills set it to zero by construction. The azimuth ϕ' is the covariant one built from the four-vectors [15], equal to $\phi_e - \phi_S$ for a massless target and differing from it by $y^2/4$ radians for a massive one — 1.4 mrad at the four bins of §3.4, 5.9 mrad at the low configuration's own spots [16] — and the 25 mrad crossing angle changes only odd harmonics of the electron azimuth because the ePIC axis is the electron beam.

The measurement itself cannot use a single state: the fit then also measures the detector's own $\cos 2\phi'$ acceptance harmonic divided by P_{zz} , and a 3% harmonic aligned with the vertical spin axis fakes an amplitude of 0.05, five to eleven times the signal. The estimator is therefore the spin-state-sorted ratio of $m = \pm 1$ -rich ($P_{zz} = +0.6$) and $m = 0$ -rich (-1.2) bunches. It cancels any acceptance common to both states bin by bin, turns relative-luminosity errors into a constant, and gives, with N the total of both states, $\delta \hat{A} = 2\sqrt{(2/N)/(P_+ - P_0)} = 0.67\times$ the single-state error if the $m = 0$ -rich fill reaches $|P_{zz}| = 1.2$. The cancellation of the acceptance, however, holds only for an acceptance common to both states: for two fills the bias is $(\epsilon_2^+ - \epsilon_2^0)/(P_+ - P_0)$ at any luminosity split, so a 10^{-3} difference of the $\cos 2\phi'$ harmonic fakes 5.6×10^{-4} , 6–13% of the projected amplitudes and 2–6 one-year statistical errors. Bunch-by-bunch alternation, or 10^{-4} stability of the harmonic between fills, is thus a requirement of the measurement rather than a refinement; with it, a relative-luminosity error between the states enters only as a constant. The generator-level drivers behind the figures of §5.1 and §6 carry the same ratio estimator behind a switch, so the single-state projections shown here and their two-fill counterparts come from one code path rather than from two implementations of the same algebra.

3.4 • Acceptance and binning

Rates use per-nucleon luminosity, F_2^A/A , and scattered-electron cuts $|\eta| \leq 3.5$, $E' \geq 0.5$ GeV, $0.01 \leq y \leq 0.95$ and $W^2 \geq 10$ GeV², which accept 5.2×10^9 events in one year (Figure 1). The four displayed (x, Q^2) super-bins are selected by statistical significance within Q^2 bands — an unconstrained ranking concentrates at the lowest Q^2 , whereas the Q^2 lever arm provides the consistency check against power-suppressed contributions — and each merges 3×3 cells of the 40×30 analysis grid (3×2 for the two $Q^2 = 1.14$ GeV² bins, clipped at the grid's $Q^2 = 1$ GeV² edge). They sit at $(x, Q^2) = (0.028, 1.14)$, $(0.011, 1.14)$, $(0.071, 3.13)$ and $(0.141, 14.3$ GeV²), i.e. at $y = 0.010, 0.025, 0.011$ and 0.025 : the amplitude grows as $(1 - y)/y^2$, so the sensitivity lives at the lowest y the detector can reconstruct, which is what makes the kinematic reconstruction of §5.2 the central detector question.

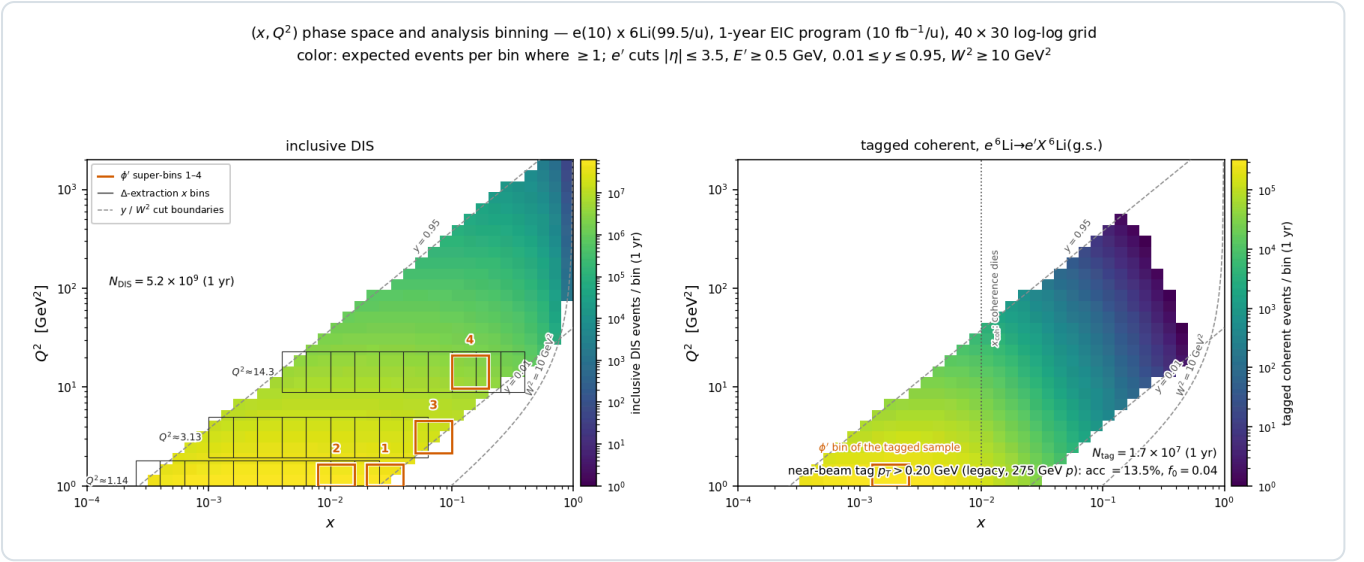


Figure 1 — phase-space coverage and analysis binning at $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV/u})$, one EIC year, 40×30 log-log grid; colour: expected events per bin where ≥ 1 ; dashed: y and W^2 cut boundaries. **Left, inclusive DIS**: boxes 1–4 are the ϕ' super-bins of Figure 2; grey combs are the three Q^2 slices ($Q^2 \times [1/1.6, 1.6]$) whose merged x -bin pairs give the data points of the Δ extraction (Figure 3) and, for the $Q^2 = 1.14 \text{ GeV}^2$ slice alone, of the amplitude-versus- x panel of Figure 2, retained where $N \geq 10^3$ and $\delta A \leq 8 \times 10^{-3}$. **Right, tagged coherent channel**: the same map times $f_{\text{coh}}(x)$ of §6.2 and the 0.20 GeV reference tag of §6.1, almost entirely at $x < 0.03$; the box is the super-bin of Figure 5(d).

4 • The Δ model

The default model, A, is an ansatz normalized to the bag moment applied to the per-nucleon Δ of ${}^6\text{Li}$:

$$\Delta_A(x, Q^2) = \frac{1}{3} A \alpha_s(Q^2) F_1(x, Q^2) \frac{x^a(1-x)^b}{N_{\text{peak}}}, \quad \int_0^1 x \Delta(x, \langle Q^2 \rangle) dx = -0.012 \alpha_s \Rightarrow A = -0.292$$

with $(a, b) = (0.7, 3)$. The factor $1/3$ restricts the exotic glue to the polarized deuteron-like pair while the whole-nucleus P_{zz} remains a separate factor, so nothing is double counted. It is a counting fraction and not a polarization: the cluster picture that fixes the vector polarization of that same pair — 0.811 of the ${}^6\text{Li}$ spin, carried by $g_1({}^6\text{Li})$ since 2026-08-29 — is a rank-1 quantity, while Δ , like b_1 , is rank 2, where the α -d wave function transfers 0.92 rather than 0.87 (Report 0, Table 3), so no upside on this factor follows from it; whether Δ should carry that rank-2 transfer as b_1 does — $1/3 \rightarrow 0.307$, an 8% reduction — is left open in the Δ -model registry, and no projection here assumes it. The constraint is imposed at the rate-weighted $\langle Q^2 \rangle = 4.47 \text{ GeV}^2$ of the accepted phase space, and because F_1 evolves the realized moment varies by a few per cent across the plotted Q^2 slices; α_s comes from the CT18NLO table. Model B imposes the same constraint without the F_1 factor and yields amplitudes a factor 3–56 lower at the four bins (20–56 in the $Q^2 = 1.14 \text{ GeV}^2$ bins, 3–8 at the higher Q^2); the A-versus-B spread is the dominant ansatz systematic, and both curves appear on every plot. Porting a single-hadron bag moment to a per-nucleon nuclear structure function has no precedent, so normalization and x dependence are both scenario choices, bracketed by model B from below and possible nuclear enhancement from above.

5 • Inclusive projections

5.1 • Truth level

Figure 2 shows the ϕ' pseudo-data in the four super-bins with one-year statistics and the extracted amplitude versus x along the $Q^2 = 1.14 \text{ GeV}^2$ slice. The injected amplitudes are 7.4, 4.4, 9.5 and 9.5×10^{-3} against one-year errors of 1.7, 1.4, 2.7 and 4.5×10^{-4} — 43, 31, 35 and 21σ — and ten-year errors of 0.55, 0.45, 0.87 and 1.4×10^{-4} ; the fitted values agree with the injected ones within 1.5 standard errors. Under the moment constraint the measurement is not a null test: the model-A-versus-B spread is resolved by the x dependence of the amplitude, and a flat amplitude of 3×10^{-3} per unit P_{zz} would remain a 3–11 σ effect per bin even at $P_{zz} = 0.3$. At ten-year precision the statistical errors sit below the polarimetry scale systematic (5% of an amplitude of 10^{-2} is 5×10^{-4} , 3×10^{-4} in the optimistic 3% case; §7), so the ten-year measurement is systematics-limited.

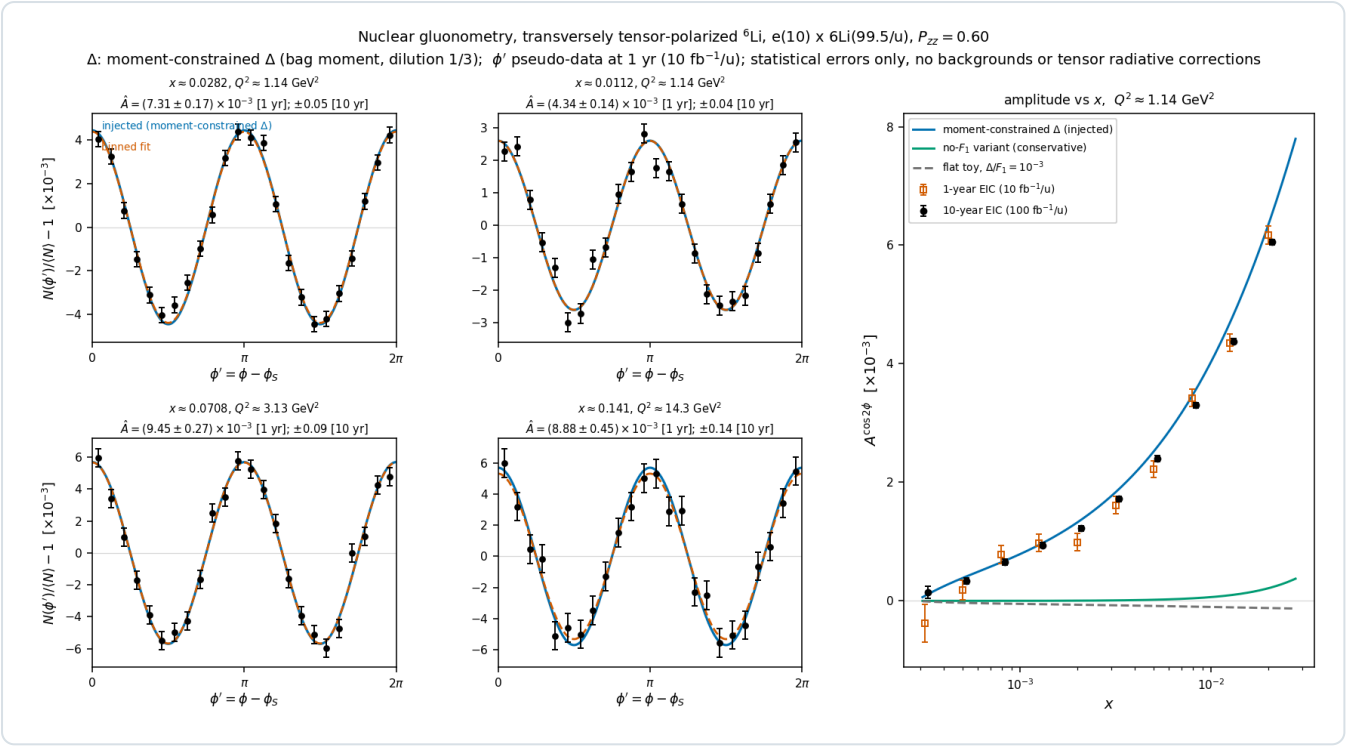


Figure 2 — inclusive gluonometry at $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV/u})$, $P_{zz} = 0.60$, model A with dilution 1/3. Left 2x2: ϕ' pseudo-data with one-year statistical errors in the four super-bins ($Q^2 = 1.1 \rightarrow 14 \text{ GeV}^2$, $x = 0.011\text{--}0.14$); injected curve (blue), binned fit (vermillion); annotations give δA for both programmes. Right: extracted amplitude versus x with one-year (open) and ten-year (filled) errors against the model A, model B and flat- 10^{-3} curves.

Figure 3 unfolds the same pseudo-measurements to the structure function, $x\Delta(x, Q^2)$ per nucleon at the three Q^2 slices of Figure 1, with independent one- and ten-year draws and the model bin-centering factor $K = \Delta_{\text{model}}(x_c, Q_c^2)/A_{\text{model, bin}}$; the area under each curve is the ported bag moment ($\div 3$ dilution). The best bins reach $\Delta = -0.135 \pm 0.003$ at $(x, Q^2) = (0.02, 1.14)$, -0.070 ± 0.002 at $(0.05, 3.13)$ and -0.0047 ± 0.0004 at $(0.32, 14.3)$ in one year, i.e. 2–9% relative, and three times better in ten.

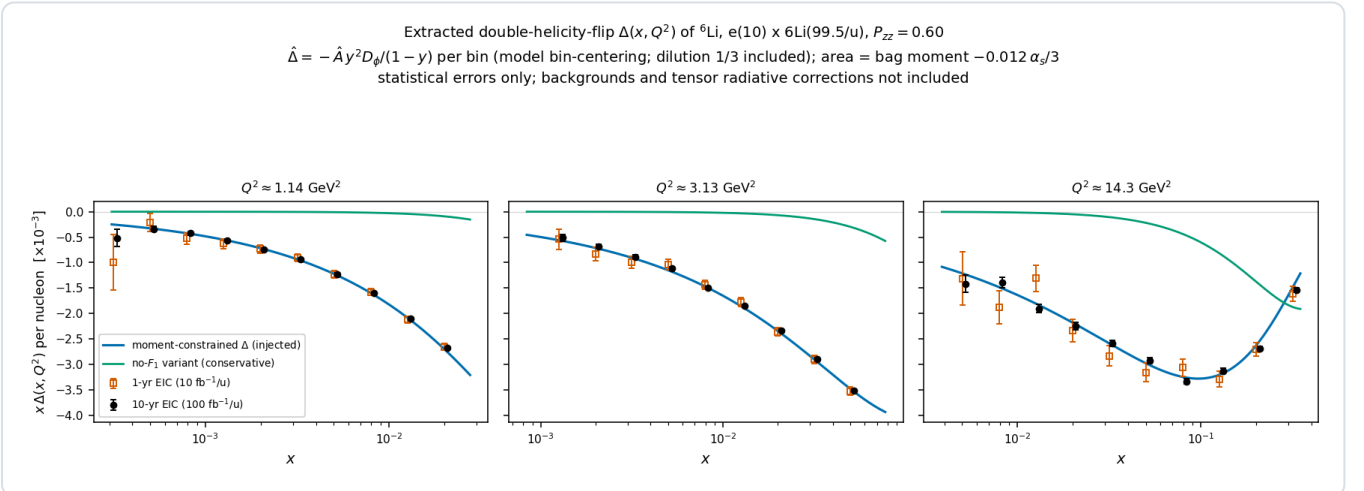


Figure 3 — the extracted $\Delta(x, Q^2)$: the amplitudes of Figure 2 unfolded to $x\Delta$ per nucleon at the three highest-significance Q^2 slices (model bin-centering, $\delta\Delta = \delta A \cdot |\partial\Delta/\partial A|$), with independent one-year (open) and ten-year (filled) pseudo-measurements against the model A and model B curves.

5.2 • Reconstructed level

Two of the four super-bins sit at $y \approx 0.01$, where the electron method alone gives $\delta y/y \approx 1.1\text{--}1.2$ and the bin is not reconstructible, and the other two at $y \approx 0.025$, where it gives 0.46; x must come from a hadronic-method y . The reconstructed-level pseudo-experiments of the companion note [16] pass every event through the lab frame, EMCal energy and track-angle smearing, the head-on transformation, the electron-method Q^2 , a hadronic y , the mixed-method x , the analysis cuts on reconstructed quantities, an η -dependent electron identification efficiency and the covariant ϕ' , then form

the spin-state ratio of the two fills with a $3\% \cos 2\varphi'$ and $2\% \cos \varphi'$ acceptance harmonic and a 10^{-3} relative-luminosity offset switched on. Table 1 gives the result with the documented 25% Gaussian stand-in for the hadronic y — the ePIC kinematic-fit study's smearing and the ATHENA value at $y \approx 0.01$ [17] — and with the y of the PYTHIA 8 final state through the hadron-side detector response (Σ method, 50 MeV of calorimeter noise), the hadronic scale calibrated from the same simulation as an analysis would, in bins of the reconstructed (x, Q^2) [16].

SUPER-BIN (x, Q^2)	25% HADRONIC-Y STAND-IN			PYTHIA 8 FINAL STATE, Σ METHOD			$\Delta\hat{A}$ (1 YR)	$\Delta\hat{A}$ (10 YR)
	PURITY	EFFICIENCY	D	$\Delta y/y$	PURITY	D	[10^{-4}]	[10^{-5}]
0.028, 1.14	0.66	0.42	0.92	0.32	0.56	1.00	1.2	3.7
0.011, 1.14	0.63	0.59	0.99	0.22	0.59	1.06	0.9	2.9
0.071, 3.13	0.69	0.38	0.90	0.29	0.64	0.95	1.6	5.2
0.141, 14.3	0.69	0.66	0.96	0.15	0.75	0.98	2.9	9.3

Table 1 — the four super-bins at the reconstructed level (mid configuration, spin-state ratio of the $+0.6 / -1.2$ fills). Purity: fraction of the reconstructed bin's rate that originates in the true bin; efficiency: fraction of the true-bin rate reconstructed in the bin after electron identification; D: rate-weighted, φ' -diluted amplitude of the events in the reconstructed bin over that of the true bin; $\delta y/y$: 68% half-width of the Σ -method resolution at the spot. The PYTHIA columns have the hadronic scale calibrated in reconstructed (x, Q^2) bins; uncalibrated they read purity 0.42 / 0.53 / 0.49 / 0.68 and $D = 0.79 / 0.84 / 0.83 / 0.95$. The statistical errors are the same for both hadronic- y models to within 7%; the two $\delta\hat{A}$ columns are the stand-in's.

Three conclusions follow. The fit closes on the response-folded truth within the precision of the test: the eight pulls against the reconstructed-bin truth lie within 2.3 standard errors while the acceptance harmonic and the luminosity offset are on, and the $\sin 2\varphi'$ coefficients are consistent with zero. The errors are 0.60–0.72 of the single-state truth-level values over both hadronic- y models, and 0.60–0.67 with the stand-in y alone, essentially the $1.5\times$ gain of the $m = 0$ -rich fill on its own: the reconstruction efficiency of 0.38–0.66 is measured against the truth bin before the analysis cuts and costs the projection no statistics, migration into the reconstructed bin leaving it with 0.97–1.25 $\times$ the truth-level rate — presuming the source delivers $m = 0$ -rich bunches at $|p_{zz}| = 1.2$ with the same purity as the ± 1 fill at 0.6. And the real final state costs purity rather than statistics: only 80–92% of the hadronic $E - p_z$ sum is within the acceptance and above threshold at the four bins, 70–85% through the full response — the rest escapes forward beyond the calorimeters (7–17%, the target-fragmentation side of a $W \approx 6$ –10 GeV system) or lies below threshold — a 13–27% scale bias on y_Σ that an analysis calibrates from its simulation at the reconstructed point. Calibrated, the purities are 0.56–0.75 and the reconstructed-bin amplitude within 7% of the true-bin value (Table 1); uncalibrated they would be 0.42–0.68 and 5–21% low, and a 1% residual scale error moves $\hat{A}$ by 0.1–0.6% [16]. The noise floor decides the resolution below $y \approx 0.02$ — at the mid configuration the Σ -method $\delta y/y$ at $y = 0.005$ runs $0.22 \rightarrow 0.33 \rightarrow 0.54 \rightarrow 1.01$ for $0 \rightarrow 25 \rightarrow 50 \rightarrow 100$ MeV of calorimeter noise on a hadronic $E - p_z$ sum of 0.10 GeV there, against 0.2–0.5 GeV at the four bins themselves, a regime no published ePIC study covers. Each configuration selects its own four bins: at 5×40.8 and 18×137.5 they resolve to $0.39 / 0.23 / 0.24 / 0.11$ and $0.23 / 0.19 / 0.21 / 0.18$, so the low- x bins improve with beam energy (the hadronic $E - p_z = 2E_y$ grows with it) while the (0.141, 14.3 GeV²) bin degrades, $0.11 \rightarrow 0.15 \rightarrow 0.18$, and is best measured at the low configuration. Unfolded to Δ , the best reconstructed bins reach statistical errors $\delta\Delta = 2.4$ (2.3) $\times 10^{-3}$ at $Q^2 = 1.14$ GeV² and 1.2 (1.1) $\times 10^{-3}$ at 3.13 GeV² in one year with the stand-in (PYTHIA) y , at purities 0.56–0.57 (0.44–0.50).

5.3 • Bin centering

Correcting the reconstructed amplitude to a point value of Δ with the model's own K is circular. Because the amplitude is exactly linear in Δ , the response can instead be applied as a linear operator and a $\Delta(x)$ shape fitted through it per Q^2 slice on a bounded grid. Correcting the same pseudo-data with either of two deliberately wrong priors, the model dependence of K at the four super-bins falls from $(-4.2, +8.0, -5.6, +4.9)\%$ to $(-1.5, -1.8, -0.3, +0.5)\%$ with the model B prior and from $(+8.6, +2.5, +7.3, +3.1)\%$ to $(+4.6, +2.9, +1.8, -0.9)\%$ with the flat toy, and over the plotted bins of Figure 3 from up to 29% to $\leq 6\%$. The Δ error bar then carries four terms — statistics, the shape fit, the response Monte Carlo (0.1–1.0% per plotted bin) and the spread over priors the shape family cannot absorb (1–8%) — of which the prior spread dominates in the best-measured bins, where the one-year statistics are $\lesssim 4\%$, and statistics elsewhere.

6 · The coherent channel

6.1 · Detecting the intact recoil

In coherent diffraction the recoil retains the beam's magnetic rigidity — $A/Z = 2$ exactly, and the diffractive momentum loss $x_p < 0.01$ changes it by less than 1% — and emerges at $\theta = p_T/p \lesssim 1$ mrad, inside the beam. Of the far-forward system at IP6 — the B0 spectrometer, the off-momentum detectors, the Roman Pots and the zero-degree calorimeter, with the acceptance windows of Refs. [18,20] — only the pots can see such a recoil, and only where it clears both the 10σ beam envelope and the pots' own aperture. Both are constraints on the *angle* at the interaction point, so with an exponential t distribution of slope B ($\approx 50 \text{ GeV}^{-2}$, §6.2) the tagged fraction is

$$\varepsilon_{\text{tag}} = \exp(-B p_{T,\text{cut}}^2), \quad p_{T,\text{cut}} = \theta_{\text{cut}} A p_u, \quad \theta_{\text{cut}} = \max(10\sigma_\theta, \theta_{\text{aperture}})$$

an exponential in the square of the beam divergence times the beam momentum. Table 2 evaluates it with $B = 50 \text{ GeV}^{-2}$ at the three ${}^6\text{Li}$ configurations, using the Yellow Report's divergences per configuration and optics — a γ -matched ion has the proton's $\beta\gamma$ and hence the proton's divergence at equal normalized emittance, and pays a factor $\sqrt{2}$ only where it is rigidity-capped, i.e. at the top configuration [13] — and the pots' aperture for an intact ${}^6\text{Li}$, measured by tracking one through the ePIC far-forward geometry: the boundary is $|\theta_x| \gtrsim 2.50 / 1.51 / 0.53$ mrad in the $5 \times 41 / 10 \times 100 / 18 \times 275$ optics against $|\theta_y| \gtrsim 0.92\text{--}2.12$ mrad, and no vertical acceptance at all at 5×41 because the transport images an interaction-point angle onto the pot plane with a horizontal lever $R_{12} = 19.24 / 21.25 / 29.97$ m against a vertical R_{34} of $4.56 / 3.35 / 2.93$ m — a factor 4.2, 6.3 and 10.2 — the 5×41 entry measured on 2026-08-29 through a scratch geometry with the four Roman-pot section offsets zeroed, the real 29.6 mm insertion there sitting at 6.49 mrad on that lever — the figure the code carries and Figure 4's first panel prints — past the 5 mrad outer bound at which the far-forward routing ends, which is what shuts the vertical plane [16]; the scan was made at the proton rigidity of each optics and is applied in angle to the γ -matched beam, which assumes the far-forward magnets scale with the stored rigidity.

${}^6\text{Li}$ CONFIGURATION	P_{ION}	Σ_θ H/V, HIGH ACCEPTANCE [MRAD]	ε_{TAG} , 10Σ ENVELOPE (CIRCULAR)	POT APERTURE $\theta_x $	ε_{TAG} , APERTURE	CIRCULAR 1/E POINT: Σ_θ , B^* FACTOR
5×40.8	245 GeV	220 / 380	5×10^{-7}	2.50 mrad	9×10^{-10}	58 μrad , $\times 14.5$
10×99.5	597 GeV	180 / 180	8×10^{-26}	1.51 mrad	2×10^{-19}	24 μrad , $\times 58$
18×137.5	825 GeV	92 / 92	4×10^{-13}	0.53 mrad	1×10^{-5}	17 μrad , $\times 29$

Table 2 — the coherent tag at IP6 with published inputs. Divergences: Yellow Report Table 10.1 — its strong-hadron-cooling values, start-of-store under the injection-cooling baseline (Report 3 §3.2) — with the γ -matched species step [13]; the ε_{tag} columns are evaluated on the unrounded divergences — 91.69 μrad at the top configuration, where the rounded 92 would return 3×10^{-13} ; at 41 GeV the high-acceptance and high-divergence optics are identical, the only configuration where the Yellow Report offers no separate high-acceptance option. Envelope: the circular $\exp(-B p_{T,\text{cut}}^2)$ of the displayed formula at $10\sigma_h$, kept as the analytic published-inputs estimate; the rectangular $10(\sigma_h, \sigma_v)$ cutout the analysis applies gives $7.2 \times 10^{-8} / 1.2 \times 10^{-26} / 7.8 \times 10^{-14}$, and $7.2 \times 10^{-8} / 6.2 \times 10^{-27} / 7.1 \times 10^{-14}$ with the vertical side at the larger of silicon and envelope, which is how the pots move (Report 4 Table 1 [21]); applying the same rule to the horizontal side — per axis the recoil must clear the larger of envelope and silicon — gives $9.4 \times 10^{-10} / 6.2 \times 10^{-27} / 7.1 \times 10^{-14}$, the silicon binding at 5×40.8 and the machine at the other two, and that is the coherent tag at the published optics the rest of the series quotes (Report 3 Table 6 [13]). Aperture: the rectangular $|\theta_x| / |\theta_y|$ cutout integrated over the two-dimensional Gaussian, not the circular $\exp(-B p_T^2)$ of the displayed formula; the aperture column is the re-measurement of 2026-08-28 in the current ePIC geometry (`epic-main` at git 9aaa2969, 16 mm modules and per-energy insertions), one event per scan point, no beam envelope, on 0.05 mrad angular ladders at four azimuths; the column is each band threshold (48 / 32 / 16 mm) over its own measured lever, which the first accepted angle of the ladder reproduces to 1–4% — 0.55 mrad against the 0.53 the 16 mm block gives at 18×275 ; the 2.0 / 1.35 / 1.03 mrad it carried before were measured in a September-2024 layout. The 5×41 entry stands on the ePIC baseline proton lattice for that ring setting; the $Z/A = 0.5$ file at the ${}^6\text{Li}$ fill point gives 1.61 mrad instead — 48 mm / 29.81 m, the threshold the code applies, against a first hit at 1.60 mrad on $+x$ and 1.70 on $-x$ — a $\times 0.64$ systematic on that row alone. Circular 1/e point: the divergence that puts a circular 10σ envelope at $p_T = 1/\sqrt{B} = 0.14 \text{ GeV}$, where $L \times \varepsilon_{\text{tag}}$ is maximal for an isotropic de-squeeze of both planes and $\varepsilon_{\text{tag}} = 1/e$, and the β^* increase over the high-acceptance optics it implies. It is not the

tagging optics this report proposes: Figure 4 prices the planar-pot, horizontal-only version, whose optimum is $\beta^*_x \times 46 / 164 / 89$ with an effective $\sigma_{\theta,x}$ of $36 / 19 / 12 \mu\text{rad}$.

The conclusion is unambiguous: with any published optics the coherent intact- ${}^6\text{Li}$ channel does not exist at IP6, at any energy — the largest entry in the table is 10^{-5} — although which of the two constraints delivers the verdict changed hands with the re-measurement: at 5×40.8 the silicon now binds, at 2.50 mrad against a 2.20 mrad envelope, and at 18×137.5 the beam does, the aperture having halved to 0.53 mrad against an envelope of 0.92. It exists under two conditions. A *tagging optics*, with β^* raised until the envelope sits at $p_T = 1/\sqrt{B}$, where luminosity times acceptance is maximal, gives $\epsilon_{\text{tag}} = 1/e$ at every configuration; the $1/e$ column of Table 2 is the circular-isotropic derivation, and priced for planar pots below the optimum is a horizontal-only de-squeeze, within the range of the forward-physics optics demonstrated at the LHC — its size, and the approach it then demands of the pots against a silicon edge at 2.50 / 1.51 / 0.53 mrad, are derived below; the two levers are multiplicative and the second is the near-beam question of Report 4 [21]. Alternatively, the *secondary focus* of the second interaction region lets pots at $z = 44.0/45.5$ m reach $p_T \approx 0$, with published intact-recoil efficiencies of 47% (d), 32% (${}^3\text{He}$), 29% (${}^4\text{He}$) and 17.8% (${}^7\text{Li}$) at top energy, rising steeply at lower energies [22]; no ${}^6\text{Li}$ case is published and our interpolation is $\approx 20\%$.

Pricing the tagging optics. The β^* lever is a machine lattice setting, not a far-forward one — it is exactly what separates the Yellow Report's high-acceptance optics from its high-divergence optics — and it is paid for in luminosity: $\sigma_{\theta} \propto 1/\sqrt{\beta^*}$ while $L \propto 1/\beta^*$ per plane, which the Yellow Report's own optics pairs reproduce to 5% (10.0 against $3.14 \times 10^{33} \text{ cm}^{-2}\text{s}^{-1}$ at 275 GeV in the 1160-bunch row, for a divergence ratio 119/65) [13]. Because a recoil escapes through the horizontal gap between the pots or through the vertical one, only the plane it escapes through needs the tagging β^* : Figure 4 therefore de-squeezes the horizontal plane alone, holds the vertical at its high-acceptance value and pays $L/L_{\text{HA}} = 1/\sqrt{r}$ rather than $1/r$, with the envelope treated as the planar pots see it — a rectangle $10\sigma_h \times 10\sigma_v$ in angle, the dispersion at the pots ($D \approx 0.3$ m with the Yellow Report $\Delta p/p$) added to the horizontal side — for pots either following the envelope or fixed at the aperture measured today. With the pots fixed, the acceptance saturates over β^* at 1.2×10^{-5} at the top configuration and at 9.4×10^{-10} and 2.0×10^{-19} at the two lower ones — the analytic fraction outside the measured rectangle, which the Monte-Carlo chain of Report 4 §2.1, carrying the divergence smearing of the measured angle, returns as 1.5×10^{-5} there [21]. Either way the far-forward side must move with the optics. With the pots following, the yield peaks at $r = \beta^*_x/\beta^*_{x,\text{HA}} = 46, 164$ and 89 at the three configurations, where $\epsilon = 0.37, 0.25$ and 0.33 , the horizontal envelope sits at 0.36, 0.19 and 0.12 mrad (0.09–0.11 GeV) and the luminosity at $1/6.8, 1/12.8$ and $1/9.5$ of the high-acceptance value; de-squeezing both planes, the naive reading of Table 2, would cost $1/25$ – $1/71$ for a fifth to a quarter of the yield. Taking the placeholder of §3.1, $10 \text{ fb}^{-1}/\text{u}$ per year, as the high-acceptance luminosity, the tagging optics yields $2.4 \times 10^6, 2.4 \times 10^6$ and 6.1×10^6 tagged events per year — $2.5\times, 6.9\times$ and $3.9\times$ below the 0.20 GeV reference tag at that luminosity — and the best super-bin of §6.2 reaches 5σ floors of 1.7%, 2.3% and 1.6% per unit P_{zz} per year. In the optics' own window, $\langle |t| \rangle = 0.033$ – 0.039 GeV^2 , the shape term of §6.3 is 3.3–3.9% per unit P_{zz} — a $9\sigma, 8\sigma$ and 11σ measurement per year on that super-bin floor — and on the same floor a 1% exotic-glue term reaches 5σ in 3.0, 5.5 and 2.6 years, the square of the floor over the amplitude. These are counting statements for the best (x, Q^2) super-bin alone, which holds 19, 10 and 9% of the tagged sample; the seven- $|t|$ -bin fit of §6.3, which uses the whole tagged sample with no x binning, measures the same flat term to $\delta a_e = 0.00121, 0.00111$ and 0.00074 per year. The top configuration is the most productive at equal luminosity, with four times the coherent events of 5×41 . If the second interaction region delivered the same luminosity as the high-acceptance optics, its secondary focus at our interpolated $\approx 20\%$ would still be worth $4\times, 10\times$ and $6\times$ the IP6 optimum in yield (Figure 4, dash-dotted line); the factor scales with $L_{\text{IR8}}/L_{\text{HA}}$, which is not published, and the published efficiencies rise toward lower energies [22]. Two assumptions carry these numbers and are stated as such: the electron β^* must be raised in step to keep the spot sizes matched, and the far-forward transport of the de-squeezed lattice must remain parallel-to-point at the pots ($R_{11}\sigma^* \ll R_{12}\sigma_{\theta}$, as in TOTEM's high- β^* optics) for the envelope to be $10R_{12}\sigma_{\theta}$ at all — an assumption at the published optics as well, R_{11} being unmeasured there too (Report 3 §7, Table 9), and the first thing to establish for a de-squeezed one.

A lithium tagging optics at IP6, priced: horizontal β^* de-squeezed, vertical at high acceptance; $\sigma_\theta \propto 1/\sqrt{\beta^*}$, $L \propto 1/\beta^*$ per plane, parallel-to-point transport assumed
 $B = 50 \text{ GeV}^{-2}$, $P_{zz} = 0.60$; yields per year at $10 \text{ fb}^{-1}/\text{u}$ in the high-acceptance optics

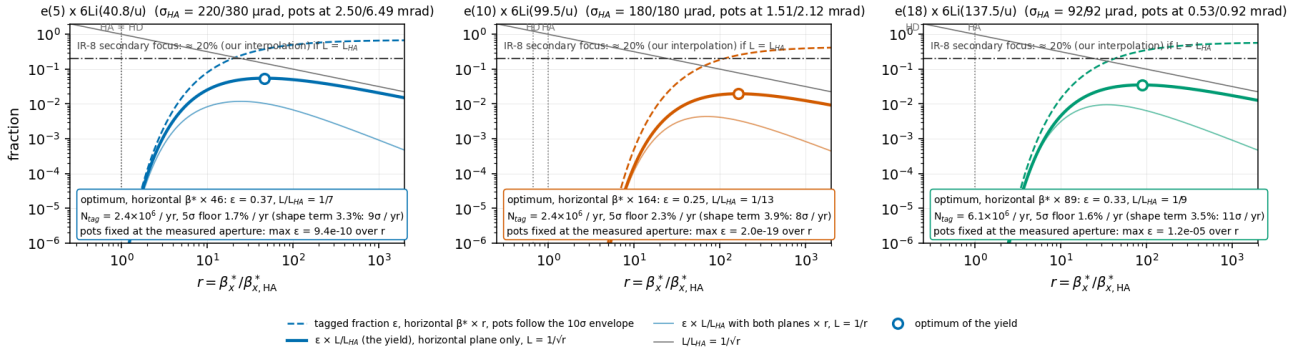


Figure 4 — a lithium tagging optics at IP6, priced. Per ${}^6\text{Li}$ configuration, against $r = \beta_x^*/\beta_{x,HA}^*$ with the vertical plane held at high acceptance: the tagged fraction ϵ for pots following the 10σ envelope (dashed); the luminosity $L/L_{HA} = 1/\sqrt{r}$ (grey); their product, the tagged yield relative to the high-acceptance luminosity (thick), with the optimum marked and the yield, best-bin 5σ floor and shape-term significance per year annotated; and the yield if both planes were de-squeezed by r (thin). The envelope is a rectangle $10\sigma_h \times 10\sigma_v$ in angle with the dispersive term added horizontally, which is why the optimum sits far above the circular $1/e$ point of Table 2; $B = 50 \text{ GeV}^{-2}$. With the pots fixed at the aperture measured in the ePIC geometry ϵ saturates over r at 1.2×10^{-5} at 18×137.5 and at 9.4×10^{-10} and 2.0×10^{-19} at 5×40.8 and 10×99.5 ; each panel states its own maximum rather than drawing it, and its title the measured cutout, whose 5×40.8 vertical entry, 6.49 mrad , is the shut plane's nominal half-width. The dash-dotted line is the IR-8 secondary focus at our interpolated $\approx 20\%$, drawn as if the second IR delivered the high-acceptance luminosity; HA and HD mark the Yellow Report optics.

The reference tag of the projections. Figures 1 and 5 and §6.2 are drawn for a near-beam envelope of 0.20 GeV — the documented limit of coherent ${}^3\text{He}$ tagging at IP6 [22], for which $\epsilon_{\text{tag}} = 13.5\%$ ($9\text{--}20\%$ for $B \in 40\text{--}60 \text{ GeV}^{-2}$) — at the high-acceptance luminosity, which no IP6 optics provides: conservative in the tagged fraction against the $25\text{--}37\%$ of the tagging optics and the $\approx 20\%$ of IR-8, but $2.5\times$, $6.9\times$ and $3.9\times$ above what the tagging optics yields at the three configurations. They stand as the physics case for a tag, not as an IP6 projection.

The configuration we propose. A lithium tagging optics at $18 \times 137.5 \text{ GeV/u}$ with the horizontal β^* raised $\approx 89\times$ over the high-acceptance setting, the vertical plane untouched, and the Roman Pots inserted horizontally to $\approx 0.12 \text{ mrad}$ (3.52 mm at the pot plane if the measured lever of 29.97 m survived the de-squeeze; the de-squeezed transfer matrix is one of the inputs requested below) — or, at the lowest energy, $5 \times 40.8 \text{ GeV/u}$ with $\beta_x^* \approx 46\times$ ($\approx 42 \text{ m}$ for a high-acceptance β_x^* of 0.9 m) and pots at 0.36 mrad , 6.99 mm ; the middle configuration asks 0.19 mrad and 4.08 mm . Against the 48 , 32 and 16 mm at which the current geometry holds the horizontal inner edge of the silicon (Report 3 §5; the vertical plane needs no move, the envelope asking 17.3 , 6.0 and 2.7 mm of pots that stand at 29.6 , 7.1 and 2.7) that is an approach a factor 6.9 , 7.9 and 4.5 closer — the envelope widths of Report 4 §2 — at the three configurations, and it is the half of the lever the β^* de-squeeze cannot supply by itself — run as a separate running configuration, as the Yellow Report foresees its high-acceptance optics, for a share of the year with which the yields above scale (Table 4, row 8) — the size of that share is a programme decision rather than a measurement, and is priced in plans/07 WP2, which puts the remainder of the year at the high-acceptance optics — while the inclusive measurement, blind to the far-forward acceptance, sets no optics requirement of its own and keeps accumulating through the de-squeezed stores at their $1/6.8\text{--}1/12.8$ of the high-acceptance luminosity. It is a change to the interaction-region lattice, on the electron side as well, and to the pot insertion; whether a horizontal de-squeeze of that size is compatible with the crab cavities and keeps the far-forward transport parallel-to-point is part of what is asked. From the machine it needs the lithium divergence and the high-acceptance β^* per configuration and the far-forward transfer matrix at the pot planes for the de-squeezed lattice; from the detector, the pot approach and the charge identification below; and its cost — a factor $6.8\text{--}12.8$ in luminosity for the duration of the store — is the number to weigh against a second interaction region. Whichever provides the tag, one degeneracy remains: a Roman-Pot hit measures A/Z , and an intact ${}^6\text{Li}$, an α and a d at the same velocity share trajectory and timing. The second-detector design answers it with a Z^2 -scaling Cherenkov counter behind the pots [19]; for IP6 no EIC document addresses charge discrimination in the pots, though the AC-LGAD readout digitizes an 8-bit charge over four planes — an α fake rate of 2.3×10^{-5} at 95% ${}^6\text{Li}$ efficiency, if the readout's unpublished full-scale charge range accommodates the ${}^6\text{Li}$ deposit — and the $\alpha + d$ pair

separates by hit topology, the partner deuteron landing on a pot in 84% of the events in which an α fakes the tag at the tagging optics [21].

6.2 • Rate model

$$\sigma_{\text{coh}} = f_{\text{coh}}(x) \sigma_{\text{DIS}}, \quad f_{\text{coh}}(x) = \frac{f_0}{1 + (x/x_{\text{coh}})^2}, \quad f_0 = 0.04 \text{ [0.02–0.08]}$$

No diffractive prediction exists for a light nucleus (the lightest published case is Ca), so the f_0 band sits below HERA ep diffraction (10–15% of DIS [23]) and heavy-nucleus saturation estimates (20–25% [24]), a loosely bound $A = 6$ nucleus taking the smaller coherent share, and spans a factor 2 either side of 0.04; parameterizing f_{coh} in Bjorken x rather than x_p is optimistic for high-mass diffraction, which the band is meant to cover. The scale $x_{\text{coh}} = 0.01$ follows the diffractive practice $x_p < 0.01$ of §6.1 rather than geometry, and sits a factor two inside the geometric point: the coherence length $l_c = 0.105 \text{ fm}/x_p$ falls below the ${}^6\text{Li}$ diameter ($\approx 5 \text{ fm}$) only near $x_p \approx 0.02$, and the slope $B = 50 \text{ GeV}^{-2}$ (band 40–60) follows from $B = \langle r^2 \rangle / 3$ with the matter radius (the charge radius gives ≈ 57 [25]; gluonic distributions may be wider [26]), valid below the form-factor minimum at $|t| \approx 0.31 \text{ GeV}^2$ [27,28]. One year produces 1.2×10^8 coherent events, of which the 0.20 GeV tag keeps 1.7×10^7 (1.7×10^8 in ten years); the best super-bin, $x \in [0.0013, 0.0025]$, $Q^2 \in [1, 1.7] \text{ GeV}^2$, holds 1.75×10^6 and gives $\delta \hat{A} = 1.8 \times 10^{-3}$ (one year) and 0.6×10^{-3} (ten years) on the amplitude per unit P_{zz} , i.e. 5σ floors at 0.9% and 0.3%; at $P_{zz} = 0.6$ the visible modulations are $0.6 \times$ these.

6.3 • The modulation and its two mechanisms

The $\cos 2\phi'$ modulation of the tagged yield carries two mechanisms. The *deformation* term is scaled from the polarized-deuteron colour-glass calculation of Mäntysaari et al. [10], the only published calculation of polarization-dependent azimuthal coefficients in coherent diffraction on any polarized nucleus (all forward citations checked; the group's subsequent work is unpolarized [29,30]). Their coefficients show a_2 growing linearly in $|t|$ with $a_2(m=0) \approx -2a_2(m=\pm 1)$ — the wave-function-symmetry relation, exact as $|t| \rightarrow 0$ and missed by 12% at the top of the analysis window by the digitized points of Figure 5(b) — from the 6% D-wave modulating the diffractive slope. For a fill with populations p_m the ensemble coefficient is proportional to P_{zz} :

$$a_2(t; P_{zz}) = -\frac{P_{zz}}{4} \varepsilon_{B0} B |t|, \quad \langle a_2 \rangle_{\text{tag}} = a_2(p_{T,\text{cut}}^2 + 1/B) = 0.036 \text{ [0.018–0.059]}$$

Two conventions matter. The anchor's Eq. (9) expands $d^2\sigma/d\Phi d|t| \propto 1 + 2\Sigma_n a_n e^{in\Phi}$ with Φ the azimuth of the recoil relative to the polarization axis in the γ^*d frame, so the $\cos 2\Phi$ coefficient is $2a_2$: the $\langle a_2 \rangle$ quoted here — 0.036 at $P_{zz} = 0.6$, i.e. 0.060 per unit P_{zz} , the footing of the 5σ floors — and injected in Figure 5(d) understates the deformation modulation by a factor two, and the coherent numbers of this paper are conservative by $\times 2$. And the modulation is one of the *recoil* azimuth $\phi_t - \phi_S$, not of the electron azimuth used as the axis of Figure 5(d); the reconstructed-level analysis fits both azimuths [16].

$\varepsilon_{B0} \equiv \Delta B_0/B$ is scaled from the deuteron value of 0.21 by two indicators that disagree: the relative quadrupole deformation $Q/\langle r^2 \rangle$ of ${}^6\text{Li}$ (0.012) is about one fifth of the deuteron's (0.063) and opposite in sign, while the α -d asymptotic D/S ratio is disputed — a transfer measurement gives $\eta = +0.0003(9)$, consistent with zero, older sub-Coulomb analyses favour $\eta \approx -0.01$, and the ab initio no-core shell model with continuum gives -0.027 , the deuteron's own 0.0256 in magnitude [31]. We adopt $\varepsilon_{B0} \in (-0.04\text{--}0.13)$ with default -0.08 , leaving near-zero deformation open; the D/S indicator alone would allow up to the deuteron's -0.21 , and the 5σ floors cover detection or exclusion anywhere in that range. The band's width is genuine theoretical uncertainty: even exact few-body calculations miss the ${}^6\text{Li}$ quadrupole cancellation by a factor ≈ 3 [32], and the gluonic deformation need not share the cancellation of the charge sector. How far that premise carries the default is arithmetic rather than an adjective: read as a charge-sector deformation, $\varepsilon_{B0} = -0.08$ at $B = 50 \text{ GeV}^{-2}$ implies a ${}^6\text{Li}$ charge quadrupole of -0.9345 fm^2 , $11.6 \times$ the measured -0.0806 fm^2 and $1.5 \times$ the α -d model's -0.615 , and inverting the measured moment and the model in turn gives a charge-sector band of $-(0.0069\text{--}0.0527)$, whose top row is the only part of it the band adopted here contains. The sign flip relative to the deuteron is a testable prediction ($Q({}^6\text{Li}) < 0$), and tensor modulations of comparable size are established at these $|t|$: the elastic e-d $T_{20} = -(0.1\text{--}0.3)$ [33] and HERMES $A_{zz}(d) \approx -1\%$ at $x \approx 0.01$ [34], the latter attributed to coherent double scattering [35–37]. The flat gluon-transversity term is

separately bounded at 3×10^{-3} – 10^{-2} (at $P_{zz} = 1$) from the bare lattice ϕ transversity [6] divided by the $\sim 10 \times$ NPLQCD deuteron suppression [7], with the maximal Drell–Yan scenario [5] as the ceiling. $\langle a_2 \rangle$ averages the m-state populations with equal weights; weighting them by their m-dependent rates lowers it by up to 27%, a one-sided shift inside the ε_{B0} band.

The two mechanisms separate by shape: deformation is linear in t , sign-flipped relative to the deuteron and expected to weaken toward small x_p (JIMWLK evolution washes it out [30]), while exotic glue is flat in t and persists at small x , where it connects to the elliptic-gluon programme [38]. In a two-component fit $a_2(t) = c_{\text{def}} \cdot |t| + a_g$ per x_p bin, the ten-year floor of 0.3% per unit P_{zz} lies at or below the flat term's band and an order of magnitude below the deformation term, so both components are expected to be resolved for the 0.20 GeV tag. Incoherent breakup is m-state-blind in this mechanism [10]: it dilutes the modulation but cannot mimic it. And deformation-driven modulations, the polarized-deuteron imaging signal, are bounded on ${}^6\text{Li}$ by its quadrupole moment, $Q = -0.0806(6) \text{ fm}^2 \approx Q({}^7\text{Li})/50$ [39]: a sizable flat-in- $t \cos 2\phi$ on ${}^6\text{Li}$ cannot be a shape effect, and the mechanisms entangled for the deuteron separate in the ${}^6\text{Li}/d$ comparison — the null test unavailable to the deuteron.

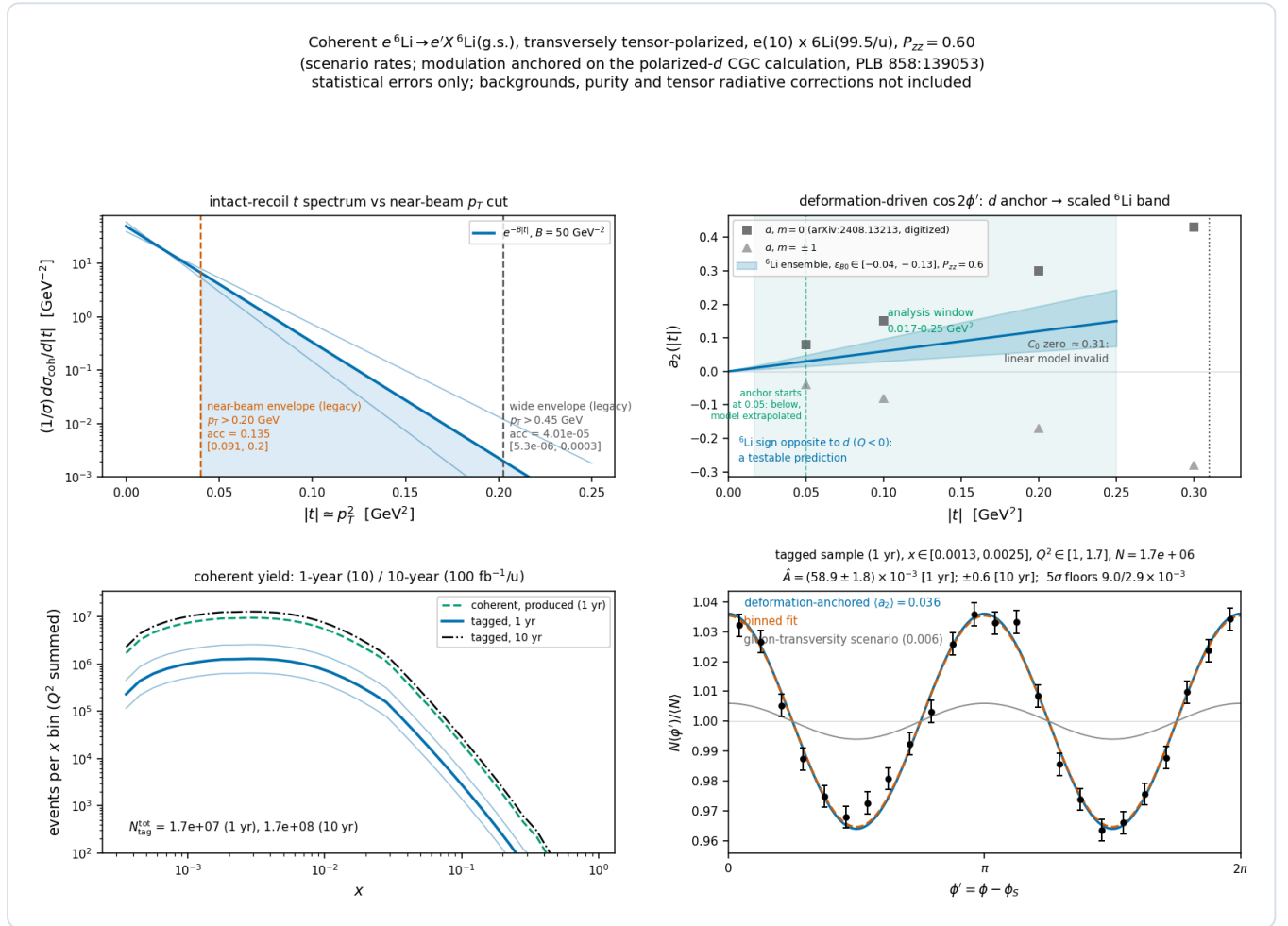


Figure 5 — the coherent channel $e {}^6\text{Li} \rightarrow e' X {}^6\text{Li}(\text{g.s.})$ at $e(10 \text{ GeV}) \times {}^6\text{Li}(99.5 \text{ GeV}/u)$, $P_{zz} = 0.60$, for a 0.20 GeV near-beam envelope; the four panels are titled rather than lettered and run (a) top left, (b) top right, (c) bottom left, (d) bottom right. (a) Recoil t spectrum with the slope band and the 0.20 and 0.45 GeV cuts (13.5% and 4×10^{-5}); both envelopes, which the panel marks *legacy*, are the proton optics the two cuts were first derived from — the 0.20 and 0.45 GeV cuts of a 275 GeV proton — neither of which is an available ${}^6\text{Li}$ optics ($\$6.1$). (b) Digitized deuteron a_2 per m state [10] against the scaled ${}^6\text{Li}$ ensemble band (sign flip predicted), with the analysis window $0.017\text{--}0.25 \text{ GeV}^2$ — the reconstructed binning of money plot 6R (Report 2 §5.2), whose lower edge is a choice of the analysis above the tagging-optics aperture floor of $0.0079\text{--}0.0132 \text{ GeV}^2$ and not the floor itself — the marker at $|t| = 0.05 \text{ GeV}^2$ where the digitized anchor begins, below which the band, and the a_t truth of the four reconstructed bins whose t_{ref} falls there, are the linear model extrapolated, and the form-factor-zero exclusion. (c) Tagged yield versus x for both programmes with the f_0 band. (d) ϕ' pseudo-data (one-year statistics) at $\langle a_2 \rangle = 0.036$ with the flat gluon-transversity scenario (grey); extraction $\hat{A} \pm 1.8 \times 10^{-3}$ (one year), $\pm 0.6 \times 10^{-3}$ (ten years) per unit P_{zz} .

That the analysis closes once a tag exists has been demonstrated with the reconstructed-level chain [16]: the recoil is transported to the pots with divergence smearing and an aperture, the spin-state ratio is formed in the two azimuths and fitted with an acceptance-weighted template basis in which the pot geometry's own $\langle \cos 2\beta \rangle$ cancels, and reflection-forbidden sin terms serve as null columns. At the tagging optics of §6.1 at 5×40.8 GeV/u — the Yellow Report divergence with the horizontal β^* de-squeezed, pots following the 0.36×3.8 mrad envelope, tagged fraction 36%, 2.3×10^6 tagged events per year at 1/6.8 of the high-acceptance luminosity, 5.5×10^5 in the lowest of seven $|t|$ bins over $0.017\text{--}0.25$ GeV² — the fit recovers the deformation coefficient $a_t = 0.0418 \pm 0.0063, 0.0593 \pm 0.0052, 0.0714 \pm 0.0052, 0.0973 \pm 0.0037, 0.1209 \pm 0.0050$ and 0.1535 ± 0.0089 against 0.046, 0.060, 0.073, 0.092, 0.121 and 0.150 injected — differing from the injected values by at most 1.5% over two hundred draws in the five bins that hold more than 10^5 recoils and by 2.5% in the sixth, which holds 5.3×10^4 — in both cases under half that bin's one-year error (Report 2 §7), and, in the seventh, with 1.3×10^4 events, 0.1166 ± 0.0186 against 0.183: the low-count bias of the bin-wise ratio, which the ten-year sample removes at 0.1799 ± 0.0060 and which is now removed outright by the acceptance-profiled likelihood that carries the same templates, returning 0.1796 ± 0.0192 on that same draw and a mean of 0.1814 over twenty one-year pseudo-experiments against 0.183 injected, where the ratio's is 0.1211 — resolves the pot roll to 15 mrad through the null columns in the $0.05\text{--}0.08$ GeV² bin and 44 mrad in the lowest, and recovers the flat term in that lowest bin to ± 0.0022 : a single pseudo-experiment returns 0.0099, and the twenty-experiment ensemble a mean of 0.0105 with a spread of 0.0018 against 0.010 injected, so the estimator is unbiased at these counts (Report 2 Table 5). Taken together the seven bins measure the flat term to $\delta a_e = 0.00121$ per year at 5×40.8 , and to 0.00111 and 0.00074 at the other two configurations, against 0.00207 for the four-bin $0.05\text{--}0.25$ GeV² window this analysis used until 2026-08-28 (Report 2 §7).

6.4 • Backgrounds

BACKGROUND	WHY IT FAKES THE TAG	HANDLE
$\alpha + d$ breakup (threshold 1.474 MeV; the 2.186 MeV 3^+ state, $\Gamma = 24$ keV, decays $\sim 100\%$ to $\alpha + d$ [40])	Both fragments at beam rigidity, on the trajectory and velocity of an intact ${}^6\text{Li}$: tracking and time-of-flight blind; only dE/dx ($\propto Z^2$: 9/4/1) or the two-hit topology separates	Roman-Pot charge discrimination (open item plans/04 #19, §6.1) and the partner-deuteron veto — given an α that fakes the tag, the deuteron is on a pot in 84% of events at the tagging optics, with no dE/dx , subject to the pot-station extent [21]. Fallback: two-component $ t $ fit (coherent $e^{-50 t }$ against a $\sim 10\times$ flatter incoherent slope; crossover at $0.05\text{--}0.2$ GeV ² , inside the window [26,41], with the three bins added below 0.05 GeV ² in 2026-08-28 lying below the crossover, so the fit now has a coherent-dominated lever arm it did not have); e+Pb benchmark 80–99% incoherent rejection [42]
Nucleon-emission breakup: $\alpha + p + n$ (3.70 MeV), ${}^3\text{He} + t$ (15.79 MeV) [40]	A fragment could pass unnoticed	$p \rightarrow$ off-momentum detectors ($R = 0.50$), $n \rightarrow$ ZDC, ${}^3\text{He} \rightarrow$ Roman-Pot window ($R = 0.75$): vetoable; inefficiency is second order
${}^6\text{Li}^*$ 3.5629 MeV (0^+ , $T = 1$, $\Gamma = 8.2$ eV) [40]	Particle-stable ($\alpha + d$ parity-forbidden); γ -decays to a real $A = 6$, $Z = 3$ track	Isoscalar exchange cannot drive $T = 0 \rightarrow 1$; the boosted $0.1\text{--}0.4$ GeV γ lands mostly in the B0 calorimeter (photons ≥ 50 MeV, $5.5\text{--}20$ mrad [20]), tail in the ZDC; the 5.366 MeV $2^+ T = 1$ state always emits a nucleon. Every $T = 1$ state is γ - or nucleon-vetoable; only $T = 0$ states feed the blind $\alpha + d$ channel
Bethe-Heitler / QED Compton	Intact nucleus by construction	Calculable (t dependence = charge form factor squared [27,28]); subtract and reuse as an acceptance candle; its charge-quadrupole $\cos 2\phi$ is $< 10^{-3}$ for $ t < 0.15$ GeV ² [28,32]
Beam-gas / halo; e^- -side fakes	Accidental beam-rigidity tracks; π / pair-symmetric e^- at high y	Non-colliding bunches, ~ 30 ps timing, vertex association; standard DIS e^- identification plus (e^- , recoil) kinematic consistency

Table 3 — backgrounds to the coherent tag, ranked. The $\alpha + d$ channel is the one that decides the purity.

7 • Systematics and assumptions

Two extraction systematics survive the spin-state ratio: the polarimetry scale $\delta P_{zz}/P_{zz} \leq 5\%$ (the working value this programme specifies, Report 2 §6; 3% is the optimistic case, reached on stored deuterons at the Nuclotron and on the

HERMES storage cell), a 1:1 normalisation on Δ common with A_{zz} and with tensor polarimetry still requiring dedicated development (Lamb-shift or Breit–Rabi-type polarimeters) [8], and radiative corrections for tensor observables, which have not been calculated. The unpolarized part is now bounded rather than assumed: collinear initial-state radiation cannot fake a $\cos 2\phi'$ — the covariant azimuth is invariant under $k \rightarrow (1-z)k$ for a massless target — it is common to the two fills and cancels in the ratio, and it leaves $x = Q_e^2/(s y_\Sigma)$ exact, moving only the Q_e^2 label of the bin by $1/(1-z)$; the migration that label carries biases Δ by +0.5 to +1.2% at the four sweet spots on the 25% Gaussian hadronic-y stand-in and +0.4 to +0.9% through the PYTHIA final state with the calibrated Σ scale ($\leq 2.9\%$ once the generator window is opened below $Q^2 = 0.15 \text{ GeV}^2$ and the low- Q^2 feed-in saturates, $\leq 0.25\%$ behind a HERA-style $E - p_z$ window the chain holds in reserve rather than applies), against the $\leq 5\%$ gate this programme set for its simulation letter (Report 2 §7). The tensor-sector correction is a different object and no unpolarized study stands in for it; the unpolarized QED tail is calculable and doubles as calibration. The detector nuisances have been measured rather than argued, holding the response fixed and varying one thing with common random numbers: a $\pm 1\%$ electron energy scale moves Δ by 0.3–1.0% at the four bins, a hadronic-resolution mismatch (0.30 generated, 0.25 corrected) by 0.5–1.4%, a 1% residual hadronic-scale error by 0.1–0.6%, all above the 0.06–0.28% Monte Carlo floor and below the bin-centering model dependence of §5.3, while an η tilt of the electron-identification efficiency ($\leq 0.03\%$) and the Yellow Report's η -dependent calorimeter table ($\leq 0.1\%$) are unresolved at that floor. In interpretation, a $\Delta\Delta$ -isobar admixture can mimic Δ at leading twist [12] and conventional double scattering generates the low- x b_1 [35–37]; the Q^2 lever arm and the ${}^6\text{Li}/d$ comparison discriminate against both. For the coherent channel, an error in the unpolarized $\cos 2(\alpha - \beta)$ harmonic u_2 reaches the flat term at first order through $a_t \delta u_2 \langle \cos 2\beta \rangle$ — a ZEUS- 1σ δu_2 [43] moves a_e by 3–9% at the tagging optics of §6.1 at 5×40.8 — 20% there under the slot geometry assumed before the aperture was measured — and 4–13% at 18×137.5 , and fitting u_1 and u_2 from the spin-averaged counts against the acceptance instead of taking them from the prior leaves 0.2–6.3% (Report 2 Table 6) — and a fill-to-fill change of the pot cutout enters through the acceptance's own $\langle \cos 2\beta \rangle$: at the tagging cutout a 10^{-3} change of the horizontal half-width moves a_t by 0.5–9.1% and a 10^{-2} one by 3.5–83.3% over the seven $|t|$ bins at 5×40.8 (0.2–31.2% and 2.4–298.6% over the three configurations), the sensitivity climbing steeply towards the lowest bin, where the horizontal edge cuts a steep part of the spectrum, and where the earlier slot geometry amplified 10^{-3} to 19% in every bin [16]. The pots' envelope must therefore be stable to 10^{-3} between the spin states — which holds the four bins above 0.05 GeV^2 to 1.9% or better, the three below it needing 10^{-4} , which leaves them within 0.9% at 5×40.8 and 3.8% across the three configurations — or the states must alternate bunch by bunch. Table 4 lists the assumptions and what a wrong one costs.

#	ASSUMPTION	IMPACT IF WRONG
1	A near-beam tag for the intact recoil exists — a tagging optics with pots that follow, or IR-8 (§6.1); the projections take a 0.20 GeV envelope at the high-acceptance luminosity, which no IP6 optics provides; the tagging-optics optimum (a 0.09–0.11 GeV horizontal envelope at 1/6.8–1/12.8 of the luminosity) yields 2.5–6.9× fewer events (§6.1)	No tag, no coherent channel; the inclusive measurement is unaffected
2	Coherent fraction $f_0 = 0.04$ [$\times 2/\div 2$], in Bjorken x (§6.2); no diffractive model for a light nucleus exists	Yields scale linearly; δA as $1/\sqrt{f_0}$
3	Bag moment ported to per-nucleon ${}^6\text{Li}$ with dilution 1/3; x shape ours (§4)	Overall Δ scale; the literature brackets both directions [7,8]; the rank-2 α -d transfer 0.92, carried by b_1 , is left open for Δ (–8% if adopted)
4	ε_{B0} scaled from the deuteron by two indicators disagreeing by $\times 3$ (§6.3)	Deformation amplitude anywhere in the band, which every plot carries
5	No event-by-event Z identification in the pots; purity from the $ t $ -shape fit (§6.4)	Z identification would improve purity; an incoherent slope model that is off grows the fit systematic
6	Linear $a_z(t)$ for $ t < 0.2 \text{ GeV}^2$ only; the form-factor zero at 0.31 GeV^2 is excluded, not modelled; and the digitized deuteron	The upper tagged window carries unmodelled structure, and the four lowest $ t $ bins — including the lowest bin of the window published until 2026-08-28 — rest on an extrapolation of the anchor rather than on an

	anchor begins at $ t = 0.05 \text{ GeV}^2$, so in the four reconstructed bins whose t_{ref} falls below it the linear shape is extrapolated past its lowest anchor point	anchor point; every bin's a_t is nonetheless quoted at a t_{ref} of 0.023–0.114 GeV^2 , inside the linear regime
7	Toy structure functions with grid backends and a placeholder R; $g_2 = g_2^{\text{WW}}$ (§3.2); the target-mass kernel on by default since 2026-08-29, the exact finite- γ TENSOR kernel available but off (§2)	Rates to $\times 1.5$; shapes are scenarios; the R shift of §3.2 applies to every Δ ; the target-mass term is a $(1 + \gamma^2)$ factor on $A_{\parallel}$ alone and reaches no Δ either way, so the flip moves nothing in this report. The tensor kernel does reach Δ , through the $O(\gamma^2)$ b_1 – b_4 leakage of §2, and is left off so that the leakage is carried as a systematic — $\leq 0.033\%$ of the amplitude here, and of the sign that cancels — rather than as a correction applied on one side of the extraction only; b_3 and b_4 , unmeasured, break the cancellation §2 rests on and would raise it to 0.052% if they were a tenth of b_2 ; a subtraction with the fit's own $\hat{\kappa}$ exists in the chain since 2026-09-06 (Report 2 §4.1), default off, and would leave 0.6 of the leakage, 0.020% here
8	All luminosity in transverse-tensor ${}^6\text{Li}$ fills at $P_{zz} = 0.6$, the $m = 0$ -rich fill at -1.2 with equal purity, alternating bunch by bunch (§3.3), and all of it at this measurement's own optics — spin state, far-forward optics (§6.1) and isotope are three layers of one run-plan choice, and every projection here takes the whole year in its own (§3.1)	N scales with the run-plan share and the amplitude with P_{zz} ; the coherent and inclusive reaches are alternatives, not a sum; without alternation the bias of §3.3
9	Impulse approximation; no final-state interactions; tensor radiative corrections unknown	FSI distorts the recoil spectrum at larger $ t $; the collinear-ISR migration is bounded at $\leq 2.9\%$ of Δ (§7, Report 2 §7), the tensor-sector correction is not
10	A calorimeter noise floor of 50 MeV on the hadronic $E - p_z$ sum, calorimeters to $ \eta = 3.7$ (70–85% of it captured at the four bins) and a hadronic scale calibrated from the simulation in reconstructed bins (§5.2) [16]	The purities of Table 1 degrade in proportion to the noise; at 100 MeV the $y \approx 0.01$ bins are lost at the mid configuration. Uncalibrated, the forward escape is a 13–27% scale bias on y and the purities fall to 0.42–0.68; a 1% residual moves $\hat{A}$ by 0.1–0.6%; the ePIC nominal reach of 4.0 recovers a third of the escape without changing $\delta y/y$

Table 4 — the assumptions behind the projections, ranked by leverage.

8 • Summary and outlook

A transversely tensor-polarized ${}^6\text{Li}$ beam at the EIC measures $\Delta(x, Q^2)$ in inclusive DIS at $21\text{--}43\sigma$ per (x, Q^2) bin in one year under the moment-constrained scenario, and at $3\text{--}11\sigma$ per bin for a flat amplitude of 3×10^{-3} per unit P_{zz} even at $P_{zz} = 0.3$, discriminating between interpretations of the bag moment by the x dependence and extracting $x\Delta(x, Q^2)$ to 2–9% in the best bins. The result survives reconstruction with the spin-state ratio and a hadronic-method y , at the price of bin purities that the calorimeter noise floor controls. The measurement makes three demands on the machine that are specific and answerable: tensor polarization delivered as alternating $m = \pm 1$ -rich and $m = 0$ -rich bunches, transverse at the interaction point; a $\cos 2\varphi'$ acceptance harmonic stable to 10^{-4} between the spin states and a pot envelope stable to 10^{-3} — 10^{-4} in the three $|t|$ bins below 0.05 GeV^2 (§7) — or the bunch-by-bunch alternation that makes any relative-luminosity error a constant; and a tensor polarimetry scale at the few-per-cent level.

The coherent channel carries the same physics with a null test the deuteron cannot offer. It is also a requirement rather than a projection: the intact ${}^6\text{Li}$ cannot be tagged at IP6 with any published optics, at any energy, and what recovers it is the tagging optics of Figure 4 or the secondary focus of the second interaction region. The tagging optics we propose (§6.1) — $18 \times 137.5 \text{ GeV/u}$ with the horizontal β^* raised $\approx 89\times$ and pots at 0.12 mrad , or $5 \times 40.8 \text{ GeV/u}$ at $\approx 46\times$ and 0.36 mrad , the vertical plane untouched, as a separate running configuration — recovers a 33–37% tag at $1/9.5\text{--}1/6.8$ of the luminosity: a 9–11 σ measurement per year of the shape term with its predicted sign flip against the deuteron, and 5σ on a 1% exotic-glue term in three years — 2.6 at the top configuration, 3.0 at the low one — both on the counting floor of the best (x, Q^2) super-bin alone, while the seven- $|t|$ -bin fit of §6.3 over the whole tagged sample measures the flat term to $\delta a_e = 0.00074$ and 0.00121 per year; a second interaction region's secondary focus would deliver 4–6 $\times$ more at the same luminosity. The highest-leverage inputs from outside the programme are, on the machine side, a light-ion optics with β^* chosen for tagging and a charge-identification study of the pot readout; on the theory side, a rerun of the IP-Glasma

polarized-deuteron setup with an α -d cluster density, which would replace both ϵ_{B0} indicators and the f_0 band at once, a lattice Δ moment for a light nucleus, and radiative corrections for tensor observables.

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Appendix A · Revision history

DATE REVISION

2026-09-15	\$3.3 records that the generator-level drivers now carry the spin-state-sorted ratio estimator behind a switch, on the run-plan pair the reconstructed-level chain uses, so that the single-state projections of §5.1 and §6 and their two-fill counterparts are produced by one code path; the published single-state figures and every number quoted from them are unchanged, and the two-fill re-derivation is recorded in plans/07 §7.1 pending the author's decision on which footing the paper quotes. §6.3 states what the adopted $\epsilon_{B0} = -0.08$ implies about the ^6Li charge quadrupole — -0.9345 fm^2 , $11.6\times$ the measured moment — and the charge-sector band $-(0.0069-0.0527)$ that inverting the measured and model quadrupoles gives, so that the distance between the no-cancellation premise and the charge sector is a number; the adopted band and every projection riding on it are unchanged (plans/04 #27 is the author's).
2026-09-06	\$2 and Table 4 row 7 restate the $O(\gamma^2)$ tensor leakage as the b_2 term: 99.96% of it is b_2 rather than b_1 , in the same combination as the ϕ' -independent constant κ of the ratio fit, so the correction is subtractable in situ from the same fills — $-L D_\phi \hat{\kappa}$ with $L = A_{\text{leak}}/\kappa$ a pure kinematic function — and not, as both sections had said, against the A_{zz} those fills measure. The subtraction is implemented and tested in the reconstruction chain (Report 2 §4.1) and, like the exact tensor kernel it corrects, is off by default, so no number of this report moves; what it would leave is the unmeasured b_3 , b_4 band, 0.6 of the leakage — $\leq 0.020\%$ of the amplitude at this configuration and $\leq 0.066\%$ at 5×40.8 .
2026-09-02	Consistency review across the five reports and the documents behind them (docs/consistency_review_2026-09-02.md); 48 of its findings land here, and they are roundings, scopes and missing qualifiers re-derived from the current libraries rather than a new run: no conclusion of this report moves. The significances are the exact quotients rather than those the printed digits give — 43 , 31 , 35σ at the four sweet spots and 21 – 43σ in the abstract and §8 (F163) — the P_{zz} scale systematic is the programme's own $\delta P_{zz}/P_{zz} \leq 5\%$, worth 5×10^{-4} on a 10^{-2} amplitude, with the 3% of the storage-cell measurements carried as the optimistic case (F004), and §2 attributes the b_1 , b_2 sign to this programme's transcription of [2], which the physics case [9] displays as well (F002), and states what the 2026-08-29 sign flip does reach: the 10^{-3} of the rate the b_1 term carries, one printed digit at the third sweet spot (F153). §3 states the vertical spin axis as an assumption on the transport into the interaction point (F172), the $\phi - \phi'$ difference as $\gamma^2/4$ radians — 1.4 mrad at the four bins, 5.9 mrad at the low configuration — in place of a 5 mrad bound (F060), the second and fourth spots at $y = 0.025$ (F033), the R-model shift as $+18$, $+18$ (F190), and, in Figure 2's caption, which combs feed which panel and which section fixes f_{coh} and the reference tag (F181, F057). In §5.2 the reconstructed-over-truth error ratio is 0.60 – 0.72 over both hadronic- y models and 0.60 – 0.67 with the stand-in y alone, and it is the $1.5\times$ gain of the $m = 0$ -rich fill on its own, Table 1's efficiency being measured against the truth bin before the analysis cuts (F066, F213, F165); that table's $\delta\hat{A}$ columns are labelled the stand-in's (F031), the noise-floor scan is quoted at the hadronic sum of 0.10 GeV where it is run (F166), and the PYTHIA $\delta\Delta$ at 3.13 GeV^2 is 1.1×10^{-3} (F191). §5.3's closure spots read $(-4.2, +8.0, -5.6, +4.9)\%$ and $(+8.6, +2.5, +7.3, +3.1)\%$ (F069) and the response-Monte-Carlo term of the published Δ bar 0.1 – 1.0% per plotted bin, the value Report 2 carries for that bar, in place of the 0.15 – 1.5% of the bare stand-in fold, which is not the bar any report shows (F010, F128). In §6.1 Table 2's caption names its inputs — the Yellow Report's strong-hadron-cooling, start-of-store divergences, the unrounded $91.69 \mu\text{rad}$ the top row's ϵ_{tag} is evaluated on, and the circular envelope column beside the rectangular cutout the analysis applies (C001, F167, F036) — the 6.49 mrad of the shut vertical plane is glossed where the text denies that acceptance and where Figure 4's panel titles print it (F185, F182), the fixed-pot maxima are given per configuration, 1.2×10^{-5} at 18×137.5 against 9.4×10^{-10} and 2.0×10^{-19} (F182), the pot approach is measured against the 48 , 32 and 16 mm horizontal edges of the current silicon, a factor 6.9 , 7.9 and 4.5 (F001), the remainder of the year is placed at the high-acceptance optics with the inclusive measurement accumulating through the de-squeezed stores as well (F126), the top configuration yields 6.1×10^6 tagged events a year (F179), the parallel-to-point transport is an assumption at the published optics too (F135), the 0.20 GeV envelope rather than the 13.5% it implies carries the ^3He tagging reference (F125), the luminosity pair is the 1160 -bunch row (F188), the acceptance windows cite [18,20] (F178), and the charge-discrimination sentence carries the 2.3×10^{-5} α fake rate at 95% ^6Li efficiency, conditional on the readout's unpublished charge range, and the 84% partner-deuteron veto (F041). §6.2 puts the f_0 band below both of its anchors, a factor 2 either side of 0.04 (F168), and takes $x_{\text{coh}} = 0.01$ from diffractive practice rather than geometry, the coherence length crossing the ^6Li diameter only near $x_p \approx 0.02$ (F019, F169). §6.3 marks $a_z(m = 0) \approx -2a_z(m = \pm 1)$ exact only as $ t \rightarrow 0$ (F187), the ratio estimator's departure from the injected values as $\leq 1.5\%$ and 2.5% , under half the bin error in both cases (F170), and, in Figure 5's caption, the panel order and the legacy proton envelopes the first panel marks (F186, F180). §7 splits the stability requirement — the acceptance harmonic to 10^{-4} , the pot envelope to 10^{-3} and to 10^{-4} in the three $ t $ bins below 0.05 GeV^2 (F071) — and carries the 0.2 – 6.3% the in-situ (u_1 , u_2) fit leaves (F171), 0.3 – 1.0% for the electron energy scale (F061) and a third rather than a quarter of the forward escape recovered at the nominal η reach (F199); Table 3's charge-discrimination row gains the partner-deuteron veto and points at plans/04 #19 (F041, F177). The years to 5σ and the shape-term significance of §1, §6.1

and §8 are identified as the counting floor of the best (x, Q^2) super-bin, with the seven-[t]-bin δa_e of the whole tagged sample beside them (F007). Figures 3 and 5 regenerated with their titles set on three lines and the bag-moment area restored to Figure 3's (F184, F183). The revision rows below are corrected where the same review found them stale: the lever triple and the zero-insertion geometry it was read off, the row the 2026-08-29 entry points at, the pot edges it quotes, the one digit the tensor-sign flip moved, and the claim that the references are numbered in citation order (F011, F056, F001, F153, F176). Wording suggestions of the same review applied: Table 2's last column is labelled the circular 1/e point it is, against the planar-pot optimum Figure 4 prices (F072), and its caption gains the cut the analysis applies on both axes, $9.4 \times 10^{-10} / 6.2 \times 10^{-27} / 7.1 \times 10^{-14}$, and the band-threshold-over-lever construction behind the aperture column (F027, F077); the fixed-pot ceiling of §6.1 and Figure 4 reads as a saturation and is reconciled with the divergence-smeared 1.5×10^{-5} of Report 4 (F079); §7 labels the ISR bias by chain (F063) and the δu_z leakage by configuration (F070); §4 and Table 4 row 3 state the rank-2 transfer as an open question for Δ (F030), row 1 quotes $2.5\text{--}6.9 \times$ (F059), row 6 adds the $0.023\text{--}0.114 \text{ GeV}^2 t_{\text{ref}}$ range (F058), and the retired likelihood triple below is labelled with the four-bin window it was measured on (F055).

2026-08-29	Verification pass over the round. The horizontal-to-vertical lever ratio is quoted per configuration, 4.2 at 5×41, 6.3 at 10×100 and 10.2 at 18×275, in place of a single factor 9 that is none of them; the 5×41 vertical lever, which the 2026-08-28 ladder could not regress through the real 29.6 mm insertion, was read off a zero-insertion scratch geometry the same day, $R_{34} = 4.56 \text{ m}$, and §6.1 carries the full triple. The $Z/A = 0.5$ lattice edge is quoted as the 1.61 mrad the code applies (48 mm / 29.81 m), with the measured first hit at 1.60 on $+x$ and 1.70 on $-x$. The closure bound of §5.2 is stated as 2.3 standard errors, the value Report 2 also carries. The de-squeeze factors and the pot approach angles are stated once, where they are derived; the cross-reference in Figure 5's caption points at Report 2 §5.2, where money plot 6R is tabulated, and the last revision row at §6.1 rather than §5.2. The dateline names Report 4. §6.1 and Figure 4's caption carry the fixed-pot ceiling the 2026-08-28 aperture re-measurement leaves: with the pots held at the measured silicon the tagged fraction peaks at 1.2×10^{-5} rather than staying below 10^{-6}, the horizontal edge at 18×137.5 having moved from 1.03 to 0.53 mrad; the conclusion that the far-forward side must move with the optics is unchanged, and so are the yields, which are quoted for pots that follow the envelope. The ceiling is stated as a maximum rather than as a bound, in the text and in Figure 4's own panel annotation, which now reads $\max \epsilon = 1.2 \times 10^{-5}$ over r in place of a $< 1 \times 10^{-5}$ the printed 1.23×10^{-5} does not satisfy (<code>tagging_optics.py</code>). Table 1's spot-4 one-year bar is 2.9×10^{-4}, §5.2's low-configuration resolution 0.39 and its 7R stand-in error at $Q^2 = 1.14 \text{ GeV}^2$ 2.4×10^{-3}, and the calibrated amplitude closes within 7% of the true-bin value rather than 6%. This appendix now runs newest first as Reports 2–4 do. Later the same day the per-configuration far-forward transport became the baseline in place of the single 18×275 lever, and every number in this report that stands on it moved: the de-squeeze factors are 46.5 / 164.1 / 89.3 and the luminosity $1/6.8 / 1/12.8 / 1/9.5$, the tagged yields $2.4 / 2.4 / 6.1 \times 10^6$ a year, the tagged fractions 0.37 / 0.25 / 0.33, three to six years to 5σ on a 1% exotic-glue term, and a second interaction region worth 4–10$\times$ rather than 3–8$\times$. The 2026-08-28 coherent-channel row below was restated on those levers rather than annotated, so its numbers read as the current ones and not as those of that day. §6.1 now states the pot insertion the optics asks for in millimetres — 6.99, 4.08 and 3.52 mm against the 48, 32 and 16 mm horizontal edges of the measured silicon — so the ask is a requirement list and not an open lever. §2's target-mass clause follows the generator's default, which was flipped on the same day: <code>InclusiveKernel</code> carries the exact finite-γ $A_{ }$ unless asked not to, and the $(1 + \gamma^2)$ factor it computes reaches no Δ either way. §4 no longer offers the cluster picture as a $\times 2.4$ upside on the dilution: the cluster value was settled that day, at 0.811 of the ${}^6\text{Li}$ spin, and it is the vector polarization of the deuteron-like pair, a rank-1 quantity that Δ cannot take — nor was the gap ever 2.4 once the per-nucleon convention was fixed. No number in this report moves with it. The tensor-sector sign was settled the same day, on the literature's convention ($A_{zz} = -(2/3) b_1/F_1$), which is one constant in the simulation and moves no Δ here and one error bar's last digit, §5.1's third-spot one-year δA, $2.8 \rightarrow 2.7 \times 10^{-4}$; and §2's $O(\gamma^2)$ $b_1\text{--}b_4$ leakage stopped being a bound, the exact finite-γ tensor kernel putting it at -0.033% of the amplitude at these four bins — of the sign that cancels part of a negative-Δ amplitude rather than faking one — against the $\leq 1\%$ the bound allowed. That figure was first published as -0.23% and corrected the same day, when the two finite-γ rows of the source's own Table 1 were made the anchor of the kernel: they fix the relative sign of the twist-3 Eq. (17d) channel against the leading-twist one, and it is the cancelling sign, so the leakage is Eq. (17e) almost alone rather than seven times it.
2026-08-28	Table 2's pot aperture re-measured in the current ePIC geometry (<code>epic-main</code> at git 9aaa2969; plans/09 B1, Report 4 §6): $\theta_x \geq 2.50 / 1.51 / 0.53 \text{ mrad}$ against the September-2024 $2.0 / 1.35 / 1.03$, so ϵ_{tag} at the aperture is $9 \times 10^{-10} / 2 \times 10^{-19} / 1 \times 10^{-5}$, and the transport that images an IP angle onto the pot plane is measured per configuration for the first time, $R_{12} = 19.24 / 21.25 / 29.97 \text{ m}$ against $R_{34} = - / 3.35 / 2.93 \text{ m}$, the 5×41 vertical entry filled in on 2026-08-29 (4.56 m; the row above). The conclusion of §6.1 stands — the largest entry is still 10^{-5} — but which constraint delivers it changes hands: the silicon binds at 5×40.8 and the beam at 18×137.5, the reverse of the earlier reading at both ends. The 5×41 row is conditional on which of that ring setting's two lattices a ${}^6\text{Li}$ fill runs in, worth $\times 0.64$ in the aperture. The ϵ_{tag}-envelope column and every projection of §6 are unchanged, the tagging optics depending on the pot dispersion only through D/R_{12}, whose re-measured 18×275 value lands within 0.6% of the one it was priced with.
2026-08-28	§3.1 completed with the two halves of the run-plan law it had stated only for statistical errors: a luminosity quoted as a reach in fb^{-1}/u is invariant under the share, the years needed to accumulate it go as one over it, and what is not statistics follows neither — the full one-year Δ bars of Report 2 [16] carry the spread over unfolding priors, which no luminosity shrinks. Assumption row 8 of Table 4 and the configuration proposed in §6.1 name the three layers the

share divides — spin state, far-forward optics and isotope — and plans/07 WP2 prices the options. No projection moves.

2026-08-28 The coherent channel of §6.3 and §7 restated on the seven $|t|$ bins over 0.017–0.25 GeV² adopted in Report 2 §7, which replace the four of 0.05–0.25 GeV² every earlier revision quoted. At 5×40.8 the window now holds 82% of the tagged sample instead of 28% and measures the flat exotic-glue term to $\delta a_e = 0.00121$ per year instead of 0.00207; every coherent number of §6.3 is re-measured on it, as are the two coherent systematics of §7 — a ZEUS-1 σ δu_2 moves a_e by 3–9% rather than 7–9%, and a 10^{-3} cutout change moves a_t by 0.5–9.1% rather than $\leq 0.8\%$, the added low bins being the least sensitive to the first and the most to the second, so the 10^{-3} envelope requirement is now edge-of-window dependent. Assumption row 6 of Table 4 gains the lower-side caveat that the digitized deuteron anchor begins at $|t| = 0.05$ GeV², and Figure 5(b) shades the analysis window and marks that start. No inclusive projection moves.

2026-08-28 The target-mass term of the longitudinal $A_{||}$ implemented behind `InclusiveKernel(target_mass=True)`, default off, and bounded where §2 and assumption row 7 of Table 4 had asserted it small without a number: $\gamma^2 \leq 0.0057$ at the four bins of §3.4, where the exact kernel collapses to a multiplicative $(1 + \gamma^2)$ on $A_{||}$ worth $\leq 0.6\% \rightarrow \leq 0.5\%$ at the top configuration and $\leq 2.5\%$ at the low one, whose $Q^2 = 1.14$ GeV² spot sits at $x = 0.089$ — and ≤ 0.097 anywhere the $W^2 \geq 10$ GeV² cut allows (`evgen/scripts/target_mass_bound.py`). No Δ reported here is built from $A_{||}$, and no published number moves.

2026-08-28 The unpolarized radiative correction moved from an exclusion to a bound (plans/07 WP4; Report 2 §7). §7 and assumption row 9 now carry the measured collinear-ISR migration — +0.5 to +1.2% of Δ at the four sweet spots in the published generator window and $\leq 2.9\%$ once that window is opened below $Q^2 = 0.15$ GeV², against the programme's $\leq 5\%$ gate — in place of the earlier $\leq 1.1\%$, which quoted one response seed and the truncated window. The tensor-sector correction remains uncalculated and is stated as such. §6.3 gained the acceptance-profiled likelihood of Report 2 §4.3, which removes the low-count bias of the sparsest $|t|$ bin outright (mean 0.173 ± 0.005 against 0.180 injected on the same twenty draws, where the ratio gives 0.114, on the four-bin 0.05–0.25 GeV² window then in use); the ratio remains the default and no published error bar moves.

2026-08-28 Numbers of §5.2 and §7 restated after the review of the reconstruction toolchain (Report 2 Appendix A, `docs/code_review_2026-08-28.md`): the hadronic-scale calibration is now keyed on the reconstructed (x, Q^2) rather than the event's true cell, and the PYTHIA library was regenerated without PYTHIA's default $\hat{m} \geq 4$ GeV floor, so the calibrated purities of Table 1 read 0.56 / 0.59 / 0.64 / 0.75 ($D = 1.00 / 1.06 / 0.95 / 0.98$) instead of 0.52 / 0.59 / 0.59 / 0.76; the coherent systematics of §7 are quoted at the tagging optics of §6.1 ($\delta u_2 \rightarrow 7\text{--}9\%$ on a_e , a 10^{-3} cutout change $\rightarrow \leq 0.8\%$ on a_t) instead of the superseded slot geometry. No projection of §5.1 or §6 moved.

2026-08-27 Rewritten as a paper on the γ -matched beam energies (⁶Li at 40.8 / 99.5 / 137.5 GeV/u, plans/10). All numbers restated from the current scripts: sweet spots at $x = 0.011\text{--}0.14$, amplitudes $(0.44\text{--}0.95) \times 10^{-2}$ against $\delta A = (1.4\text{--}4.5) \times 10^{-4}$, the reconstructed-level Table 1, the R-model shifts (+18, +19, +8, -4%), the bin-centering scan, 1.2×10^8 coherent events and 1.7×10^7 tagged for the 0.20 GeV envelope. The coherent channel restated against the Yellow Report divergences and the measured pot aperture (Table 2): it does not survive at IP6 with any published optics, and the figures are presented for a tagging optics or IR-8 rather than as an IP6 projection; the pre-2026-08-27 6R and measured-aperture results, computed at the superseded rigidity-scaled energies (2.7×10^6 tagged at 50 GeV/u; the low-energy survival at 20.5 GeV/u), are withdrawn and replaced by the analysis closure at a 0.73 mrad approach. Assumptions condensed to Table 4; the reference list renumbered. Later the same day: the tagging optics priced (Figure 4, `tagging_optics.py`) — luminosity $\propto 1/\beta^*$ per plane against the tagged fraction, a horizontal-only de-squeeze with the vertical plane untouched, dispersion at the pots and the parallel-to-point transport assumption stated, the pots following or fixed — and the IP6 lithium tagging configuration proposed in §6.1 with its cost.

2026-08-26 Development run 8: the PYTHIA 8 hadronic final state; the Roman-Pot aperture measured in the ePIC geometry; the published R1998 fit behind an `r_func` hook; the forward-folded bin-centering alternative.

2026-08-25 The spin-state-sorted ratio as the estimator of the measurement; the near-beam cut stated as angular; the b_1, b_2 sign convention; the inversion corrected; the $O(\gamma^2)$ b_1 leakage; far-forward parameters updated.

2026-08-24 The anchored coefficient's convention ($2a_2$); the reconstruction-chain report created as the reconstructed-level companion.

2026-08-17 Dedicated fact checks of the Sather–Schmidt moment usage and of the far-forward detection chain; phase-space figure with the analysis binning; display mathematics typeset at build time.

2026-08-11 Second audit round (formulas against the literature, implementation, references, extraction logic); money plot 7, the extracted Δ as structure-function data points, added.

2026-08-10 First version (development run 5): money plots 5 and 6 with the moment-constrained $\Delta(x, Q^2)$ and separate one-year / ten-year projections, and the physics note with a verified bibliography; first audit round the same day.

Generated from the PolarizedLithiumSim repository: `evgen/scripts/money_cos2phi.py`, `money_cos2phi_coherent.py`, `money_delta_extraction.py`, `phase_space_bins.py`; reconstructed level `money_cos2phi_reco.py` (with `--y-source hfs`, `--unfold-scan`, `--syst-scan`), `hfs_resolution.py`, `coherent_optics_scan.py`, `tagging_optics.py`, `nearbeam_aperture_scan.py`, `nearbeam_reach_gain.py`; models in `polli_fastsim/delta_models.py` and `evgen/polligen/coherent.py`; assembled by `reports/build_report.py`. Every command is in `docs/reproduction_manual.md`. All rates use scenario inputs — bands, not predictions; external numbers are verified against primary sources. Target venue: INT programme on polarized ion beams at the EIC, March 22 – April 2, 2027.